Stochastic and Sequential Regimes: Extending the Decision Quotient to Dynamic Information Sufficiency

Tristan Simas
McGill University
tristan.simas@mail.mcgill.ca

March 1, 2026

© 2026 Tristan Simas. This work is licensed under CC BY 4.0. License: https://creativecommons.org/licenses/by/4.0/
Zenodo DOI: 10.5281/zenodo.18140965

Abstract

We extend the static decision quotient framework of Paper 4 to stochastic and sequential regimes. Given a decision problem with actions A, factored state space $S = X_1 \times \cdots \times X_n$, and utility U, we ask: what complexity is required to identify sufficient coordinate sets when (a) the state is drawn from a distribution P, or (b) decisions unfold over time with transitions and observations?

Results:

- **Stochastic regime**: STOCHASTIC-SUFFICIENCY-CHECK is PP-complete via MAJSAT reduction. Tractable under product distributions, bounded support, log-concave.
- **Sequential regime**: SEQUENTIAL-SUFFICIENCY-CHECK is PSPACE-complete via TQBF reduction. Tractable under full observability, bounded horizon, tree structure.
- **Regime hierarchy**: Static $\subset$ Stochastic $\subset$ Sequential; coNP $\subset$ PP $\subset$ PSPACE.
- **Substrate cost**: The integrity-competence verdict is substrate-independent; trajectories diverge based on substrate cost κ.
- **Temporal learning**: Bayesian structure detection reduces abstention over time.
- **Temporal integrity**: Evidence-monotone retractions preserve integrity across sequences.

The reduction constructions are machine-checked in Lean 4 (3288 lines, 124 theorems).

Keywords: computational complexity, decision theory, stochastic decision problems, POMDPs, polynomial hierarchy, PSPACE

1 Introduction

Consider any decision problem $\mathcal{D}$ with actions A and factored state space $S = X_1 \times \cdots \times X_n$. A coordinate set $I \subseteq \{1, \ldots, n\}$ is *sufficient* if knowing only coordinates in I determines the optimal action:

$$s_I = s'_I \implies \mathrm{Opt}(s) = \mathrm{Opt}(s')$$

This is *coordinate sufficiency*—the formalization of *information sufficiency* (whether available information determines the optimal action) over a factored state space.

Paper 4 established that SUFFICIENCY-CHECK is coNP-complete for static decision problems (A, S, U). This paper extends that landscape to two natural extensions: stochastic and sequential regimes.

1.1 Stochastic Regime

When the state is drawn from a distribution $P : \Delta(S)$, sufficiency must account for randomness. We use the distribution-conditional definition: I is stochastically sufficient if $P(s \mid s_I) \otimes U$ determines the same optimal action as $P(s) \otimes U$. This subsumes expected-value sufficiency (where only the first moment matters) and connects to classical sufficient statistics.

Regime	Tuple	Complexity
Static	(A, S, U)	coNP-complete (Paper 4)
Stochastic	(A, S, U, P)	PP-complete
Sequential	(A, S, U, T, O)	PSPACE-complete

The PPupper bound comes from counting satisfying assignments weighted by P. The lower bound uses a MAJSAT reduction: given φ, construct uniform P over $\{0,1\}^n$ and let $U(\text{accept}, s) = \varphi(s)$, $U(\text{reject}, s) = 1/2$. Then $\emptyset$ is stochastically sufficient iff $\mathbb{E}_P[\varphi(s)] \geq 1/2$ or $< 1/2$—i.e., majority of assignments satisfy φ.

Tractable subcases: product distributions (expected utility decomposes), bounded support ($|\text{supp}(P)| \leq k$), log-concave distributions.

1.2 Sequential Regime

When decisions unfold over time with transitions $T : A \times S \to \Delta(S)$ and observations $O : S \to \Delta(\Omega)$, the problem becomes a POMDP. Sequential sufficiency asks whether observation history restricted to I-coordinates determines the optimal policy.

The PSPACEupper bound follows from Papadimitriou & Tsitsiklis (1987). The lower bound uses TQBF reduction: given $\forall x_1 \exists x_2 \forall x_3 \ldots \varphi$, construct n time steps, adversarial transitions at universal quantifiers, agent choices at existential quantifiers, terminal utility φ. Then full observation history is sequentially sufficient iff the QBF is true.

Tractable subcases: fully observable (MDP), bounded horizon, tree-structured transitions, deterministic transitions.

1.3 Substrate Cost

We extend Paper 4's integrity-competence matrix with substrate cost $\kappa(\text{cell}, \text{agentType})$. The matrix verdict is substrate-independent: $\text{verdict}(c)$ is the same for all agent types. But trajectories diverge: $\text{trajectory}(\sigma, \tau_1) \neq \text{trajectory}(\sigma, \tau_2)$ in general. This is the minimal formal hook for substrate in decision theory.

1.4 Temporal Learning

An agent facing sequential problem instances drawn from unknown distribution D can learn the structure class via Bayesian updates. As $k \to \infty$, posterior $P(C \mid I_1, \ldots, I_k) \to \delta_{C*}$ if structure classes are identifiable. This reduces the abstention frontier: $A_1 \supseteq A_2 \supseteq \ldots$.

1.5 Temporal Integrity

Claims across sequences must satisfy evidence-monotone retractions: an agent retracts a claim only when new evidence contradicts it. Retraction without evidence violates temporal integrity; refinement ($I' \subset I$) is allowed.

1.6 Main Theorems

1. **Theorem 2.6:** STOCHASTIC-SUFFICIENCY-CHECK is PP-complete via MAJSAT reduction.
2. **Theorem 3.6:** SEQUENTIAL-SUFFICIENCY-CHECK is PSPACE-complete via TQBF reduction.
3. **Propositions 4.1 and 4.2 (Regime Hierarchy):** Static $\subset$ Stochastic $\subset$ Sequential; coNP$\subset$ PP$\subset$ PSPACE.
4. **Theorem 5.2 (Substrate Independence of Verdict):** For all cells c, verdict(c) is κ-independent.
5. **Theorem 5.3 (Substrate Dependence of Trajectory):** There exist τ_1, τ_2, σ with trajectory$(\sigma, \tau_1) \neq$ trajectory(σ, τ_2).

(L: CC58, CC48, CC64, CC63)

1.7 Claim-Typing Bridges (Inherited Static Core)

1. **Typed claim admissibility (inherited static core):** report admissibility is typed by certificate class (ABSTAIN, EXACT, EPSILON(ε)).
2. **Evidence-admissibility equivalence (inherited static core):** first-class evidence-object existence is equivalent to typed report admissibility.
3. **Empty-sufficient characterization (inherited static core):** the $I = \emptyset$ query is exactly universal constancy of the decision boundary, and non-sufficiency is exactly existence of a counterexample pair.
4. **Exact requires evidence (inherited static core):** exact admissibility is equivalent to exact evidence existence.
5. **No evidence implies zero certified confidence (inherited static core):** under signal consistency, no evidence implies zero certified confidence.

[C:P4-core typed-claim chain;R:DC,AR] (L: CC75,IC7,DP2,DP5,IC9,IC17)

1.8 Paper Structure

Section 2: Stochastic decision problems, PP-completeness, tractable cases. Section 3: Sequential (POMDP) problems, PSPACE-completeness, tractable cases. Section 4: Strict hierarchy of regimes. Section 5: Substrate cost formalization and independence results. Section 6: Bayesian structure detection, abstention reduction. Section 7: Temporal integrity constraints. Section 8: Transfer conditions between regimes. Section 9: Related work.

2 Stochastic Decision Problems

We extend the static decision problem to include a probability distribution over states.

2.1 Stochastic Decision Problem

Definition 2.1 (Stochastic Decision Problem)**.** A *stochastic decision problem with coordinate structure* is a tuple $\mathcal{D}_S = (A, X_1, \ldots, X_n, U, P)$ where:

- A is a finite set of *actions*
- $X_1, \ldots, X_n$ are finite *coordinate spaces*
- $S = X_1 \times \cdots \times X_n$ is the *state space*
- $U : A \times S \to \mathbb{Q}$ is the *utility function*
- $P \in \Delta(S)$ is a *distribution over states*

Definition 2.2 (Distribution-Conditional Sufficiency)**.** A coordinate set $I \subseteq \{1, \ldots, n\}$ is *stochastically sufficient* for $\mathcal{D}_S$ if:

$$P(s \mid s_I) \otimes U \text{ determines the same optimal action as } P(s) \otimes U.$$

Equivalently, for all $s, s' \in S$ with $s_I = s'_I$:

$$\arg\max_{a \in A} \mathbb{E}_{t \sim P(\cdot \mid s_I)}[U(a,t)] = \arg\max_{a \in A} \mathbb{E}_{t \sim P}[U(a,t)].$$

Definition 2.3 (Expected-Value Sufficiency (Alternative))**.** A coordinate set I is *expected-value sufficient* if:

$$\forall a, a' : \mathbb{E}_{s \sim P}[U(a,s) \mid s_I] \geq \mathbb{E}_{s \sim P}[U(a',s) \mid s_I] \implies a \in \mathrm{Opt}_P.$$

where $\mathrm{Opt}_P = \arg\max_{a \in A} \mathbb{E}_{s \sim P}[U(a,s)]$.

Decision 2.4. STOCHASTIC-SUFFICIENCY-CHECK: Given stochastic decision problem $\mathcal{D}_S = (A, S, U, P)$ and coordinate set I, is I stochastically sufficient?

Decision 2.5. STOCHASTIC-MINIMUM-SUFFICIENT-SET: Given $\mathcal{D}_S$, find the minimal $I \subseteq \{1, \ldots, n\}$ that is stochastically sufficient.

2.2 Complexity: PP-Completeness

Theorem 2.6 (STOCHASTIC-SUFFICIENCY-CHECK is PP-complete)**.** *STOCHASTIC-SUFFICIENCY-CHECK is PP-complete.*

1. **Membership in PP**: Given $\mathcal{D}_S$ and I, we must check whether knowing s_I determines the optimal action. This reduces to comparing expected utilities conditioned on s_I versus unconditioned. For each action pair (a, a'), we compare:

   $$\mathbb{E}_{s \sim P}[U(a,s) \mid s_I] \geq \mathbb{E}_{s \sim P}[U(a',s) \mid s_I].$$

   This requires counting weighted by P, which is in PP.

2. **PP-hardness via MAJSAT reduction**: Given Boolean formula φ on n variables, construct:

   - $S = \{0,1\}^n$ (state space = assignments)
   - P = uniform distribution over S
   - $A = \{\text{accept}, \text{reject}\}$

- $U(\text{accept}, s) = \varphi(s)$, $U(\text{reject}, s) = 1/2$
- $I = \emptyset$

Then $\emptyset$ is stochastically sufficient iff the optimal action is determined without seeing any coordinates:

$$\emptyset \text{ sufficient} \iff \mathbb{E}_{s\sim P}[\varphi(s)] \geq 1/2 \text{ or } < 1/2.$$

This is exactly MAJSAT, which is PP-complete.

(L: CC58, CC55)

Proof. For membership: We reduce to a PPpredicate. For each $a \in A$, compute $\mathbb{E}_P[U(a, s)]$ and compare. The comparison involves summing over $s \in S$ weighted by $P(s)$, which is polynomial in $|S|$ and the representation of P. The decision is whether the argmax is unique or whether knowing s_I changes it—this is a majority-vote comparison, hence in PP.

For hardness: The reduction from MAJSAT constructs the uniform distribution. If φ is true on $\geq 1/2$ of assignments, the expected utility of accepting is $\geq 1/2$, making accept optimal regardless of state observation. If $< 1/2$, reject is optimal. Thus $\emptyset$ is sufficient exactly when MAJSAT is true. ■

Example 2.7 (Stochastic Weather)**.** Consider stochastic umbrella decision:

- $A = \{\text{carry}, \text{don't carry}\}$
- $S = \{\text{rain}, \text{no rain}\}$
- $P(\text{rain}) = 0.3$, $P(\text{no rain}) = 0.7$
- $U(\text{carry}, \text{rain}) = 2$, $U(\text{carry}, \text{no rain}) = -1$
- $U(\text{don't carry}, \text{rain}) = -2$, $U(\text{don't carry}, \text{no rain}) = 0$

Expected utility: $E_P[U(\text{carry})] = 0.3(2)+0.7(-1) = -0.1$, $E_P[U(\text{don't carry})] = 0.3(-2)+0.7(0) = -0.6$. So carry is optimal. If we condition on forecast information, we check whether the conditional distribution changes the argmax.

2.3 Tractable Subcases

1. **Product distributions**: $P = P_1 \otimes \cdots \otimes P_n$. Then

$$\mathbb{E}_{s\sim P}[U(a, s) \mid s_I] = \prod_{i\in I} P_i(s_i) \cdot \mathbb{E}_{s\sim P}[U(a, s) \mid s_I],$$

   so expected utility decomposes. This reduces stochastic sufficiency to static sufficiency per coordinate.

2. **Bounded support**: $|\text{supp}(P)| \leq k$. Enumeration is $O(k \cdot |A|)$, polynomial.

3. **Log-concave distributions**: For ordered state spaces, conditional expectations are monotone and computable via convex optimization.

Claim 2.8 (Tractability under product distributions)**.** If $P = P_1 \otimes \cdots \otimes P_n$ is a product distribution, then STOCHASTIC-SUFFICIENCY-CHECK reduces to checking static sufficiency on each marginal. *[C:P2.8;R:PD]*

(L: CC60)

Claim 2.9 (Tractability under bounded support)**.** If $|\text{supp}(P)| \leq k$, then STOCHASTIC-SUFFICIENCY-CHECK is solvable in $O(k \cdot |A| \cdot n)$ time. *[C:P2.9;R:BS]*

(L: CC59)

2.4 Bridge from Static

Paper 4 proved that static sufficiency does NOT imply stochastic sufficiency (CC54). A counterexample exists where I is sufficient for (A, S, U) but not for (A, S, U, P).

However, under product distributions, static sufficiency transfers:

Claim 2.10 (Static to stochastic transfer)**.** If $P = P_1 \otimes \cdots \otimes P_n$ is a product distribution and I is statically sufficient for (A, S, U), then I is stochastically sufficient for (A, S, U, P). *[C:P2.10;R:PD]*

(L: CC52)

3 Sequential Decision Problems

We further extend to sequential decision-making with transitions and observations.

3.1 Sequential Decision Problem (POMDP)

Definition 3.1 (Sequential Decision Problem)**.** A *sequential decision problem with coordinate structure* is a tuple $\mathcal{D}_{SEQ} = (A, X_1, \ldots, X_n, U, T, O)$ where:

- A is a finite set of *actions*
- $X_1, \ldots, X_n$ are finite *coordinate spaces*
- $S = X_1 \times \cdots \times X_n$ is the *state space*
- $U : A \times S \to \mathbb{Q}$ is the *stage utility*
- $T : A \times S \to \Delta(S)$ is the *transition kernel*
- $O : S \to \Delta(\Omega)$ is the *observation model*

The agent chooses a policy $\pi : (\Omega^*) \to A$ mapping observation histories to actions. Objective: maximize expected cumulative utility $\mathbb{E}\left[\sum_{t=0}^{T-1} \gamma^t U(a_t, s_t)\right]$ with discount factor $\gamma \in [0, 1)$.

Notation 3.2 (Observation History)**.** Let $\omega_t = (o_0, o_1, \ldots, o_t)$ denote the observation history up to time t. The agent's policy is $\pi(\omega_{t-1}) = a_t$.

Definition 3.3 (Sequential Sufficiency)**.** A coordinate set $I \subseteq \{1, \ldots, n\}$ is *sequentially sufficient* for $\mathcal{D}_{SEQ}$ if the optimal policy based on full observation history is the same as the optimal policy based only on coordinates in I at each time step.

Formally: let π^* be the optimal policy with full observations, and let π_I^* be the optimal policy with observations restricted to I-coordinates. Then $\pi^* = \pi_I^*$ (as functions of observation history).

Decision 3.4. SEQUENTIAL-SUFFICIENCY-CHECK: Given sequential decision problem $\mathcal{D}_{SEQ}$ and coordinate set I, is I sequentially sufficient?

Decision 3.5. SEQUENTIAL-MINIMUM-SUFFICIENT-SET: Given $\mathcal{D}_{SEQ}$, find the minimal I that is sequentially sufficient.

3.2 Complexity: PSPACE-Completeness

Theorem 3.6 (SEQUENTIAL-SUFFICIENCY-CHECK is PSPACE-complete)**.** *SEQUENTIAL-SUFFICIENCY-CHECK is PSPACE-complete.*

1. **Membership in PSPACE**: Optimal POMDP policy computation is in PSPACE(Papadimitriou & Tsitsiklis, 1987). Checking whether a coordinate set I yields the same policy involves comparing value functions, which can be done in polynomial space via alternating polynomial time.

2. **PSPACE-hardness via TQBF reduction**: Given quantified Boolean formula $\forall x_1 \exists x_2 \forall x_3 \ldots \varphi(x_1, \ldots, x_n)$, construct:

   - n time steps, one variable per step
   - At odd steps (universal quantifier): "nature" chooses the variable (adversarial transition)
   - At even steps (existential quantifier): agent chooses the variable (action)
   - Terminal utility: $\varphi(x_1, \ldots, x_n)$
   - Observation: agent sees all previous variables
   - I = full observation history

   Then full observation is sequentially sufficient iff the agent can guarantee $\varphi = \text{true}$ regardless of nature's choices, which holds iff the QBF is true.

(L: CC48, CC47)

Construction 3.7 (TQBF Reduction)**.** Given QBF $\Phi = Q_1 x_1 Q_2 x_2 \ldots Q_n x_n \ \varphi(x_1, \ldots, x_n)$ with alternating quantifiers, construct sequential problem:

- Time steps $t = 1, \ldots, n$
- State at step t encodes $(x_1, \ldots, x_t)$
- At step t: - If $Q_t = \forall$: transition is adversarial (nature chooses both next state and observation) - If $Q_t = \exists$: transition is controlled by agent action (choose x_t)
- Terminal reward: 1 if $\varphi(x_1, \ldots, x_n) = \text{true}$, 0 otherwise
- Full observation reveals all previous variables

Then: full observation is sequentially sufficient $\iff$ there exists a policy guaranteeing $\varphi = \text{true}$ $\iff$ Φ is true.

(L: CC77)

Claim 3.8 (Sequential sufficiency implies static sufficiency)**.** When the horizon $T = 1$ and transitions are deterministic, sequential sufficiency reduces to static sufficiency. *[C:P3.8;R:T=1,DET]*

(L: CC46)

3.3 Tractable Subcases

1. **Fully observable (MDP)**: O is the identity (agent sees full state). Then the problem reduces to a stochastic (or deterministic) problem per time step. Value iteration is polynomial in $|S|, |A|, T$.

2. **Bounded horizon**: $T \leq k$. The state space grows as $O(|S|^T)$, but for fixed T this is polynomial in $|S|$.

3. **Tree-structured transitions**: The transition graph has no cycles. Dynamic programming yields polynomial algorithms.

4. **Deterministic transitions**: $T(s'|a, s)$ is a point mass. The problem reduces to search over deterministic paths.

Claim 3.9 (Tractability under full observability)**.** If O is the identity (fully observable), then SEQUENTIAL-SUFFICIENCY-CHECK reduces to checking stochastic sufficiency at each time step. *[C:P3.9;R:FO]*

(L: CC50)

Claim 3.10 (Tractability under bounded horizon)**.** If $T \leq k$ (constant horizon), then SEQUENTIAL-SUFFICIENCY-CHECK is solvable in time polynomial in $|S|, |A|, k$. *[C:P3.10;R:BH]*

(L: CC49)

3.4 Bridge from Static and Stochastic

Paper 4 proved that static sufficiency does NOT transfer to sequential (horizon > 1): CC18

Similarly, stochastic sufficiency does NOT transfer to sequential in general:

Claim 3.11 (Stochastic to sequential bridge failure)**.** There exists a stochastic sufficient coordinate set I and a sequential problem where I is NOT sequentially sufficient, even when transitions are memoryless. *[C:P3.11;R:CE]*

(L: CC56)

The transfer conditions are:

- Static $\rightarrow$ Sequential: transfers iff $T = 1$ and transitions are deterministic (one-step bridge from Paper 4)

- Stochastic $\rightarrow$ Sequential: transfers iff transitions are memoryless AND observation model is regime-compatible

4 Regime Hierarchy

We establish the strict inclusion hierarchy among decision regimes.

Inclusion	Complexity Classes	Condition
Static $\subset$ Stochastic	coNP$\subset$ PP	Standard assumptions (P $\neq$ coNP)
Stochastic $\subset$ Sequential	PP$\subset$ PSPACE	Standard assumptions (P $\neq$ PP)

Claim 4.1 (Strict inclusion static $\subset$ stochastic)**.** Under standard complexity assumptions (P $\neq$ coNP), there exist stochastic decision problems that are not statically reducible. *[C:P4.1;R:P $\neq$ coNP]*

(L: P4SS1.)

Claim 4.2 (Strict inclusion stochastic $\subset$ sequential)**.** Under standard complexity assumptions (P $\neq$ PP), there exist sequential decision problems that are not stochastically reducible. *[C:P4.2;R:P $\neq$ PP]*

(L: P4SS2.)

4.1 Integrity-Resource Bound per Regime

The integrity-resource bound from Paper 4 generalizes to each regime:

1. **Static** (coNP-hard): polynomial-time agent must abstain on some instances
2. **Stochastic** (PP-hard): polynomial-time agent must abstain on MORE instances
3. **Sequential** (PSPACE-hard): polynomial-time agent must abstain on EVEN MORE instances

Formally, if $C_1 \subsetneq C_2$ (complexity classes), then the set of instances where integrity forces abstention under C_1-resources is a strict superset of those under C_2-resources:

$$\{\sigma : \text{integrity forces abstention at } C_1\} \supsetneq \{\sigma : \text{integrity forces abstention at } C_2\}$$

Claim 4.3 (Abstention frontier shifts)**.** As regime complexity increases (static $\to$ stochastic $\to$ sequential), the abstention frontier expands. *[C:P4.3;R:RG]*

(L: CC39)

4.2 Regime Detection

A meta-problem: given a problem instance, which regime is it in?

Decision 4.4. REGIME-DETECTION: Given a problem instance, is it static, stochastic, or sequential?

Conjecture: regime detection is polynomial for the represented family:

- Bounded actions $\to$ static
- Distribution present $\to$ stochastic
- Transitions/observations present $\to$ sequential

(L: CC41)

5 Substrate Cost

We extend the integrity-competence matrix from Paper 4 with substrate cost κ.

5.1 Definition

Definition 5.1 (Substrate Cost Function)**.** $\kappa : (\text{MatrixCell} \times \text{AgentType}) \to (\mathbb{R}_{\geq 0}, \leq)$

Where:

- MatrixCell = (integrity, attempted, competenceAvailable) triple from Paper 4
- AgentType = opaque type representing substrate (silicon, carbon, formal system, etc.)
- The codomain is a totally ordered cost space

Claim 5.2 (Verdict is substrate-independent)**.** For all cells c and agent types τ_1, τ_2:

$$\text{verdict}(c; \tau_1) = \text{verdict}(c; \tau_2).$$

[C:T5.2;R:RG]

(L: CC64)

Claim 5.3 (Trajectory is substrate-dependent)**.** There exist agent types τ_1, τ_2 and problem sequences σ such that:

$$\text{trajectory}(\sigma; \tau_1) \neq \text{trajectory}(\sigma; \tau_2)$$

even though $\text{verdict}(\sigma_t; \tau_1) = \text{verdict}(\sigma_t; \tau_2)$ for all t. *[C:T5.3;R:RG]*

(L: CC63)

5.2 What κ Captures

κ is a placeholder for everything the (A, S, U) tuple doesn't model:

- Memory
- Motivation
- Fatigue
- Identity
- Embodiment

We do NOT decompose κ. We acknowledge it exists, show where it enters the dynamics, and prove that the matrix verdict is κ-independent while the trajectory is κ-dependent.

Claim 5.4 (Minimal formal hook)**.** $\kappa(\text{cell}, \text{agentType})$ is the sufficient statistic of substrate for the trajectory problem. *[C:T5.4;R:RG]*

(L: CC65)

This is the clean separation: the model captures exactly the behaviorally relevant component of substrate, and nothing more.

We do NOT prove that κ captures all of subjective experience. We prove that κ captures all of subjective experience that is *decision-relevant*—i.e., that affects the agent's trajectory through the problem sequence.

5.3 Where Substrate Enters

The matrix verdict is computed from (integrity, attempted, competenceAvailable). None of these depend on substrate. The verdict is universal.

Substrate enters at exactly one point: the COST of being in a cell. This cost determines:

- Whether the agent attempts the next instance or rests
- How the agent allocates resources across a sequence
- The agent's effective learning rate (some substrates learn faster)
- The agent's abstention threshold (some substrates tolerate uncertainty better)

This connects to Paper 5 (Credibility): $\kappa = \kappa_{\text{compute}} + \kappa_{\text{communicate}}$, where $\kappa_{\text{communicate}}$ is bounded by Paper 5's cheap talk framework.

6 Temporal Learning

We formalize how an agent learns structure classes across sequential problem instances.

6.1 Setup

Agent faces a sequence of problem instances $I_1, I_2, \ldots$ drawn from an unknown distribution D over problem structures.

After solving instance I_k (or abstaining), the agent observes:

- Whether I_k belonged to a tractable subcase
- Which structure class C_k was detected (if any)
- The cost paid (substrate-dependent κ)

Claim 6.1 (Bayesian structure detection)**.** Agent maintains posterior $P(C \mid I_1, \ldots, I_k)$ over structure classes. Update rule: standard Bayesian update with likelihood $P(I_k \mid C)$. *[C:T6.1;R:RG]*

(L: CC4)

Convergence: does $P(C \mid I_1, \ldots, I_k) \to \delta_{C*}$ as $k \to \infty$?

Answer: yes, if the structure classes are identifiable (distinct likelihood functions).

Claim 6.2 (Identifiability implies convergence)**.** If structure classes C are identifiable (distinct likelihood functions), then $P(C \mid I_1, \ldots, I_k) \to \delta_{C*}$ as $k \to \infty$. *[C:P6.2;R:ID]*

(L: CC19)

6.2 PAC Learning Connection

Sample complexity of structure detection: How many instances must the agent solve before it can identify the structure class with probability $\geq 1 - \delta$ and error $\leq \varepsilon$?

Claim 6.3 (Sample complexity conjecture)**.** For the represented family (bounded actions, separable utility, tree structure), sample complexity is polynomial in $1/\varepsilon$ and $\log(1/\delta)$. *[C:J6.3;R:RG]*

This connects to PAC learning theory—the structure classes are the hypothesis space.

6.3 Learning Reduces Abstention

As the agent learns the structure class, it can apply regime-appropriate heuristics. The abstention frontier shrinks over time—not because the problem gets easier, but because the agent's model of the problem gets more accurate.

Formally: let A_k be the set of instances where integrity forces abstention at time k. Then:

$$A_1 \supseteq A_2 \supseteq \ldots \supseteq A_\infty$$

where A_∞ = instances genuinely outside all tractable subcases.

Claim 6.4 (Abstention reduction over time)**.** A_k is decreasing in k: $A_1 \supseteq A_2 \supseteq \ldots$. *[C:T6.4;R:RG]*

(L: CC68)

The agent's trajectory diverges based on κ even though verdicts are κ-independent. Different substrates learn at different rates, have different abstention thresholds, and make different trade-offs between solving and abstaining.

This connects to Paper 5: temporal learning produces costly signals. Each solved instance is a costly signal in Paper 5's sense: producing a correct solution has truth-dependent cost. The agent's track record is costly signal; self-assessment is cheap talk.

7 Temporal Integrity

We formalize consistency of claims across sequences of problem instances.

7.1 Definition

Definition 7.1 (Temporal Integrity)**.** Temporal integrity = consistency of claims across a sequence of instances.

An agent has temporal integrity if:

- Each per-instance claim satisfies Paper 4's integrity constraint
- Retractions are evidence-monotone: the agent retracts a claim only when new evidence contradicts it, never without cause
- Refinements are monotone: if the agent claims I is sufficient at time t, and later claims $I' \subset I$ is sufficient at time $t' > t$, this is a refinement (allowed). If it claims $I' \supset I$, this is a weakening that requires justification.

Decision 7.2. TEMPORAL-INTEGRITY-CHECK: Given a sequence of claims and evidence, is the sequence temporally integrity-preserving?

Claim 7.3 (Retraction with evidence preserves integrity)**.** If an agent claimed I sufficient, observed a counterexample, then retracted, this preserves temporal integrity. *[C:P7.3;R:RG]*

(L: CC70)

Claim 7.4 (Retraction without evidence violates integrity)**.** If an agent claimed I sufficient, observed no new information, then retracted, this violates temporal integrity. *[C:P7.4;R:RG]*

(L: CC71)

Claim 7.5 (Refinement strengthens integrity)**.** If an agent claimed I sufficient, discovers $I' \subset I$ also sufficient, updates claim to I'. This preserves and strengthens temporal integrity. *[C:P7.5;R:RG]*

(L: CC69)

7.2 Complexity of Temporal Integrity

Claim 7.6 (Complexity lower bound)**.** TEMPORAL-INTEGRITY-CHECK is at least as hard as the base regime's SUFFICIENCY-CHECK. *[C:T7.6;R:RG]*

Each claim in the sequence requires verification. The temporal constraint (monotonicity) adds structure but doesn't reduce worst-case complexity.

Paper 4 proves integrity forces abstention. Paper 5 proves claiming integrity is cheap talk. Together: the agent must abstain (Paper 4), cannot credibly communicate why (Paper 5, cheap talk bound), and can only escape by producing costly signals (Paper 5, Theorem 4.1)—which are exactly the mechanized proofs that Paper 4 itself provides.

8 Cross-Regime Transfer

When does a sufficiency result in regime R_1 help in regime R_2?

8.1 Transfer Conditions

1. **Static → Stochastic**: Transfers iff distribution P concentrates on a tractable static subcase. Formally: if $\mathrm{supp}(P)$ lies within a structure-detectable class C, then static sufficiency for C implies stochastic sufficiency under P. Otherwise: bridge failure (already mechanized).

2. **Static → Sequential**: Transfers iff horizon $T = 1$ and transitions are deterministic. This is the one-step bridge (already proved in Paper 4: `CC30`). For $T > 1$: bridge failure (`CC18`).

3. **Stochastic → Sequential**: Transfers iff transitions are memoryless AND observation model is regime-compatible. Formally: if $T(s'|a, s)$ depends only on the stochastically sufficient coordinates of s, then stochastic sufficiency lifts to sequential sufficiency. Otherwise: new bridge-failure witness needed.

4. **Sequential → Static**: Always degenerates. A sequential problem with $T = 1$ and deterministic transitions IS a static problem.

Claim 8.1 (Transfer static to stochastic)**.** Static sufficiency transfers to stochastic iff P concentrates on tractable subcase. *[C:T8.1;R:RG]*

(L: `CC53`)

Claim 8.2 (Transfer stochastic to sequential)**.** Stochastic sufficiency transfers to sequential iff memoryless transitions and regime-compatible observations. *[C:T8.2;R:RG]*

(L: `CC57`)

8.2 Heuristic Transfer Across Regimes

Paper 4's heuristic reusability bridge generalizes: if structure detection is polynomial in regime R_1, and the detected structure is preserved under regime extension to R_2, then the heuristic transfers.

Key question: when is structure preserved?

- Product distributions preserve separable utility
- Deterministic transitions preserve tree structure

9 Related Work

This paper extends the static decision quotient to stochastic and sequential regimes. Related work falls into several categories.

9.1 Complexity of Dynamic Decision Problems

The complexity of MDPs and POMDPs is well-established: optimal MDP planning is in P, optimal POMDP planning is PSPACE-complete [10, 8, 6]. Our contribution extends this to the meta-question of identifying sufficient coordinates in these settings.

Sufficient statistics [3] is the classical foundation. Our stochastic sufficiency definition connects to this tradition but focuses on decision-relevant information rather than statistical estimation.

9.2 Information and Signaling

Value of information [5, 11] quantifies payment for information. We address a different question: the complexity of identifying which information is sufficient for decision.

Signaling games [12, 9, 7] connect to our temporal integrity and credibility results (Paper 5). The agent must abstain (Paper 4), cannot credibly signal why (Paper 5), and can only escape via costly signals.

9.3 Feature Selection in Machine Learning

Feature selection asks which inputs predict outputs, known NP-hard in general [2, 1, 4]. Our decision-theoretic analog for sufficiency is PP-complete (stochastic) and PSPACE-complete (sequential).

9.4 Regime Complexity Hierarchy

Our hierarchy Static $\subset$ Stochastic $\subset$ Sequential with complexity classes coNP $\subset$ PP $\subset$ PSPACE extends the representation-sensitive complexity results established for MDPs/POMDPs to the meta-level sufficiency problem.

This paper integrates: (i) stochastic and sequential extensions, (ii) substrate cost formalization, (iii) temporal learning and integrity, and (iv) cross-regime transfer conditions.

10 Conclusion

This paper extends the static decision quotient to stochastic and sequential regimes.

Summary of Results

- **Stochastic sufficiency** (Section 2): STOCHASTIC-SUFFICIENCY-CHECK is PP-complete via MAJSAT reduction. Tractable under product distributions, bounded support, log-concave.
- **Sequential sufficiency** (Section 3): SEQUENTIAL-SUFFICIENCY-CHECK is PSPACE-complete via TQBF reduction. Tractable under full observability, bounded horizon, tree structure.
- **Regime hierarchy** (Section 4): Static $\subset$ Stochastic $\subset$ Sequential; coNP $\subset$ PP $\subset$ PSPACE.

- **Substrate cost** (Section 5): Verdict is κ-independent; trajectory is κ-dependent.
- **Temporal learning** (Section 6): Bayesian structure detection reduces abstention over time.
- **Temporal integrity** (Section 7): Evidence-monotone retractions preserve integrity.

These results place stochastic and sequential sufficiency at higher complexity levels than static sufficiency.

Open Questions

- **Exact transfer conditions:** Characterize when stochastic sufficiency transfers to sequential.
- **PAC learning bounds:** Prove sample complexity bounds for structure detection.
- **Temporal integrity complexity:** Tight complexity bounds for TEMPORAL-INTEGRITY-CHECK.

This work extends Paper 4's framework to dynamic settings while preserving the core insight: identifying sufficient information is computationally hard, and integrity forces abstention when resources don't match problem complexity.

11 Lean 4 Proof Listings

The complete Lean 4 formalization extends the Paper 4 artifact. The mechanization consists of 3288 lines across 29 files, with 124 theorem/lemma statements.
Handle IDs. Inline theorem metadata cites compact IDs (for example, `CC4`, `HD2`, `IC3`) instead of full theorem constants.

11.1 Lean Handle ID Map

Lean handle entry
AC1 DecisionQuotient.ClaimClosure.AtomicCircuitExports.AC1
AC3 DecisionQuotient.ClaimClosure.AtomicCircuitExports.AC3
AC4 DecisionQuotient.ClaimClosure.AtomicCircuitExports.AC4
AC5 DecisionQuotient.ClaimClosure.AtomicCircuitExports.AC5
AC6 DecisionQuotient.ClaimClosure.AtomicCircuitExports.AC6
AC8 DecisionQuotient.ClaimClosure.AtomicCircuitExports.AC8
AC9 DecisionQuotient.ClaimClosure.AtomicCircuitExports.AC9
AC11 DecisionQuotient.ClaimClosure.AtomicCircuitExports.AC11
AN1 DecisionQuotient.Physics.AssumptionNecessity.no_assumption_free_proof_of_refutable_claim
AN2 DecisionQuotient.Physics.AssumptionNecessity.countermodel_violates_some_assumption
AN3 DecisionQuotient.Physics.AssumptionNecessity.physical_claim_requires_physical_assumption
AN4 DecisionQuotient.Physics.AssumptionNecessity.physical_claim_requires_empirically_justified_physical_assumption
AN5 DecisionQuotient.Physics.AssumptionNecessity.strong_physical_no_go_meta
AQ1 DecisionQuotient.ClaimClosure.AQ1
AQ2 DecisionQuotient.ClaimClosure.AQ2

Lean handle entry (continued)

AQ3 DecisionQuotient.ClaimClosure.AQ3

AQ4 DecisionQuotient.ClaimClosure.AQ4

AQ5 DecisionQuotient.ClaimClosure.AQ5

AQ6 DecisionQuotient.ClaimClosure.AQ6

AQ7 DecisionQuotient.ClaimClosure.AQ7

AQ8 DecisionQuotient.ClaimClosure.AQ8

ARG1 PhysicalComplexity.AccessRegime.PhysicalDevice

ARG2 PhysicalComplexity.AccessRegime.AccessRegime

ARG3 PhysicalComplexity.AccessRegime.RegimeEval

ARG4 PhysicalComplexity.AccessRegime.RegimeSample

ARG5 PhysicalComplexity.AccessRegime.RegimeProof

ARG6 PhysicalComplexity.AccessRegime.RegimeWithCertificate

ARG7 PhysicalComplexity.AccessRegime.RegimeEvalOn

ARG8 PhysicalComplexity.AccessRegime.RegimeSampleOn

ARG9 PhysicalComplexity.AccessRegime.RegimeProofOn

ARG10 PhysicalComplexity.AccessRegime.RegimeWithCertificateOn

ARG11 PhysicalComplexity.AccessRegime.HardUnderEval

ARG12 PhysicalComplexity.AccessRegime.AuditableWithCertificate

ARG13 PhysicalComplexity.AccessRegime.certificate_upgrades_regime

ARG14 PhysicalComplexity.AccessRegime.certificate_upgrades_regime_on

ARG15 PhysicalComplexity.AccessRegime.physical_succinct_certification_hard

ARG16 PhysicalComplexity.AccessRegime.certificate_amortizes_hardness

ARG17 PhysicalComplexity.AccessRegime.regime_upgrade_with_certificate

ARG18 PhysicalComplexity.AccessRegime.regime_upgrade_with_certificate_on

ARG19 PhysicalComplexity.AccessRegime.AccessChannelLaw

ARG20 PhysicalComplexity.AccessRegime.FiveWayMeet

AXM1 complete_mono

AXM2 completeD_mono

AXM3 minimal_no_redundant_axes

AXM4 semantically_minimal_implies_independent

BA1 DecisionQuotient.Physics.BoundedAcquisition.BoundedRegion

BA2 DecisionQuotient.Physics.BoundedAcquisition.acquisition_rate_bound

BA3 DecisionQuotient.Physics.BoundedAcquisition.acquisitions_are_transitions

BA4 DecisionQuotient.Physics.BoundedAcquisition.one_bit_per_transition

BA5 DecisionQuotient.Physics.BoundedAcquisition.resolution_reads_sufficient

BA6 DecisionQuotient.Physics.BoundedAcquisition.srank_le_resolution_bits

BA7 DecisionQuotient.Physics.BoundedAcquisition.energy_ge_srank_cost

BA8 DecisionQuotient.Physics.BoundedAcquisition.srank_one_energy_minimum

BA9 DecisionQuotient.Physics.BoundedAcquisition.physical_grounding_bundle

BA10 DecisionQuotient.Physics.BoundedAcquisition.counting_gap_theorem

BAS1 correctness_forcing

BAS2 dof_inconsistency_potential

BAS3 dof_gt_one_inconsistent

Lean handle entry (continued)
BB1 DecisionQuotient.BayesianDQ
BB2 DecisionQuotient.BayesianDQ.certaintyGain
BB3 DecisionQuotient.dq_is_bayesian_certainty_fraction
BB4 DecisionQuotient.bayesian_dq_matches_physics_dq
BB5 DecisionQuotient.dq_derived_from_bayes
BC1 DecisionQuotient.Foundations.counting_nonneg
BC2 DecisionQuotient.Foundations.counting_total
BC3 DecisionQuotient.Foundations.counting_additive
BC4 DecisionQuotient.Foundations.bayes_from_conditional
BC5 DecisionQuotient.Foundations.entropy_contraction
BF1 DecisionQuotient.certainty_of_not_nondegenerateBelief
BF2 DecisionQuotient.nondegenerateBelief_of_uncertaintyForced
BF3 DecisionQuotient.forced_action_under_uncertainty
BF4 DecisionQuotient.bayes_update_exists_of_nondegenerateBelief
BND1 ssot_upper_bound
BND2 non_ssot_lower_bound
BND3 ssot_advantage_unbounded
BND4 ClaimClosure.arbitrary_reduction_factor
CAP1 ClaimClosure.cap_encoding_conditional
CAP2 ssot_guarantees_coherence
CAP3 non_ssot_permits_incoherence
CC1 DecisionQuotient.ClaimClosure.RegimeSimulation
CC2 DecisionQuotient.ClaimClosure.adq_ordering
CC3 DecisionQuotient.ClaimClosure.system_transfer_licensed_iff_snapshot
CC4 DecisionQuotient.ClaimClosure.anchor_sigma2p_complete_conditional
CC5 DecisionQuotient.ClaimClosure.anchor_sigma2p_reduction_core
CC6 DecisionQuotient.ClaimClosure.anchor_query_relation_false_iff
CC7 DecisionQuotient.ClaimClosure.anchor_query_relation_true_iff
CC8 DecisionQuotient.ClaimClosure.boundaryCharacterized_iff_exists_sufficient_subset
CC9 DecisionQuotient.ClaimClosure.bounded_actions_detectable
CC10 DecisionQuotient.ClaimClosure.bridge_boundary_represented_family
CC11 DecisionQuotient.ClaimClosure.bridge_failure_witness_non_one_step
CC12 DecisionQuotient.ClaimClosure.bridge_transfer_iff_one_step_class
CC13 DecisionQuotient.ClaimClosure.certified_total_bits_split_core
CC14 DecisionQuotient.ClaimClosure.cost_asymmetry_eth_conditional
CC15 DecisionQuotient.ClaimClosure.declaredBudgetSlice
CC16 DecisionQuotient.ClaimClosure.declaredRegimeFamily_complete
CC17 DecisionQuotient.ClaimClosure.declared_physics_no_universal_exact_certifier_core
CC18 DecisionQuotient.ClaimClosure.dichotomy_conditional
CC19 DecisionQuotient.ClaimClosure.epsilon_admissible_iff_raw_lt_certified_total_core
CC20 DecisionQuotient.ClaimClosure.exact_admissible_iff_raw_lt_certified_total_core
CC21 DecisionQuotient.ClaimClosure.exact_certainty_inflation_under_hardness_core
CC22 DecisionQuotient.ClaimClosure.exact_raw_eq_certified_iff_certainty_inflation_core

Lean handle entry (continued)
CC23 DecisionQuotient.ClaimClosure.exact_raw_only_of_no_exact_admissible_core
CC24 DecisionQuotient.ClaimClosure.explicit_assumptions_required_of_not_excused_core
CC25 DecisionQuotient.ClaimClosure.explicit_state_upper_core
CC26 DecisionQuotient.ClaimClosure.hard_family_all_coords_core
CC27 DecisionQuotient.ClaimClosure.horizonTwoWitness_immediate_empty_sufficient
CC28 DecisionQuotient.ClaimClosure.horizon_gt_one_bridge_can_fail_on_sufficiency
CC29 DecisionQuotient.ClaimClosure.information_barrier_opt_oracle_core
CC30 DecisionQuotient.ClaimClosure.information_barrier_state_batch_core
CC31 DecisionQuotient.ClaimClosure.information_barrier_value_entry_core
CC32 DecisionQuotient.ClaimClosure.integrity_resource_bound_for_sufficiency
CC33 DecisionQuotient.ClaimClosure.integrity_universal_applicability_core
CC34 DecisionQuotient.ClaimClosure.meta_coordinate_irrelevant_of_invariance_on_declared_slice
CC35 DecisionQuotient.ClaimClosure.meta_coordinate_not_relevant_on_declared_slice
CC36 DecisionQuotient.ClaimClosure.minsuff_collapse_core
CC37 DecisionQuotient.ClaimClosure.minsuff_collapse_to_conp_conditional
CC38 DecisionQuotient.ClaimClosure.minsuff_conp_complete_conditional
CC39 DecisionQuotient.ClaimClosure.no_auto_minimize_of_p_neq_conp
CC40 DecisionQuotient.ClaimClosure.no_exact_claim_admissible_under_hardness_core
CC41 DecisionQuotient.ClaimClosure.no_exact_claim_under_declared_assumptions_unless_excused_core
CC42 DecisionQuotient.ClaimClosure.no_exact_identifier_implies_not_boundary_characterized
CC43 DecisionQuotient.ClaimClosure.no_uncertified_exact_claim_core
CC44 DecisionQuotient.ClaimClosure.one_step_bridge
CC45 DecisionQuotient.ClaimClosure.oracle_lattice_transfer_as_regime_simulation
CC46 DecisionQuotient.ClaimClosure.physical_crossover_above_cap_core
CC47 DecisionQuotient.ClaimClosure.physical_crossover_core
CC48 DecisionQuotient.ClaimClosure.physical_crossover_hardness_core
CC49 DecisionQuotient.ClaimClosure.physical_crossover_policy_core
CC50 DecisionQuotient.ClaimClosure.process_bridge_failure_witness
CC51 DecisionQuotient.ClaimClosure.poseAnchorQuery
CC52 DecisionQuotient.ClaimClosure.pose_returns_anchor_query_object
CC53 DecisionQuotient.ClaimClosure.posed_anchor_checked_true_implies_truth
CC54 DecisionQuotient.ClaimClosure.posed_anchor_exact_claim_admissible_iff_competent
CC55 DecisionQuotient.ClaimClosure.posed_anchor_exact_claim_requires_evidence
CC56 DecisionQuotient.ClaimClosure.posed_anchor_no_competence_no_exact_claim
CC57 DecisionQuotient.ClaimClosure.posed_anchor_query_truth_iff_exists_anchor
CC58 DecisionQuotient.ClaimClosure.posed_anchor_query_truth_iff_exists_forall
CC59 DecisionQuotient.ClaimClosure.posed_anchor_signal_positive_certified_implies_admissible
CC60 DecisionQuotient.ClaimClosure.query_obstruction_boolean_corollary
CC61 DecisionQuotient.ClaimClosure.query_obstruction_finite_state_core
CC62 DecisionQuotient.ClaimClosure.regime_core_claim_proved
CC63 DecisionQuotient.ClaimClosure.regime_simulation_transfers_hardness
CC64 DecisionQuotient.ClaimClosure.reusable_heuristic_of_detectable
CC65 DecisionQuotient.ClaimClosure.selectorSufficient_not_implies_setSufficient

Lean handle entry (continued)
CC66 DecisionQuotient.ClaimClosure.separable_detectable
CC67 DecisionQuotient.ClaimClosure.snapshot_vs_process_typed_boundary
CC68 DecisionQuotient.ClaimClosure.standard_assumption_ledger_unpack
CC69 DecisionQuotient.ClaimClosure.stochastic_objective_bridge_can_fail_on_sufficiency
CC70 DecisionQuotient.ClaimClosure.subproblem_hardness_lifts_to_full
CC71 DecisionQuotient.ClaimClosure.subproblem_transfer_as_regime_simulation
CC72 DecisionQuotient.ClaimClosure.sufficiency_conp_complete_conditional
CC73 DecisionQuotient.ClaimClosure.sufficiency_conp_reduction_core
CC74 DecisionQuotient.ClaimClosure.sufficiency_iff_dq_ratio
CC75 DecisionQuotient.ClaimClosure.sufficiency_iff_projectedOptCover_eq_opt
CC76 DecisionQuotient.ClaimClosure.thermo_conservation_additive_core
CC77 DecisionQuotient.ClaimClosure.thermo_energy_carbon_lift_core
CC78 DecisionQuotient.ClaimClosure.thermo_eventual_lift_core
CC79 DecisionQuotient.ClaimClosure.thermo_hardness_bundle_core
CC80 DecisionQuotient.ClaimClosure.thermo_mandatory_cost_core
CC81 DecisionQuotient.ClaimClosure.tractable_bounded_core
CC82 DecisionQuotient.ClaimClosure.tractable_separable_core
CC83 DecisionQuotient.ClaimClosure.tractable_subcases_conditional
CC84 DecisionQuotient.ClaimClosure.tractable_tree_core
CC85 DecisionQuotient.ClaimClosure.transition_coupled_bridge_can_fail_on_sufficiency
CC86 DecisionQuotient.ClaimClosure.tree_structure_detectable
CC87 DecisionQuotient.ClaimClosure.typed_claim_admissibility_core
CC88 DecisionQuotient.ClaimClosure.typed_static_class_completeness
CC89 DecisionQuotient.ClaimClosure.universal_solver_framing_core
CCC1 DecisionQuotient.CC.anchor_sigma2p_complete_conditional
CCC2 DecisionQuotient.CC.cost_asymmetry_eth_conditional
CCC3 DecisionQuotient.CC.dichotomy_conditional
CCC4 DecisionQuotient.CC.minsuff_collapse_to_conp_conditional
CCC5 DecisionQuotient.CC.minsuff_conp_complete_conditional
CCC6 DecisionQuotient.CC.sufficiency_conp_complete_conditional
CCC7 DecisionQuotient.CC.tractable_subcases_conditional
CF1 DecisionQuotient.Physics.ConstraintForcing.laws_not_determined_of_parameter_separation
CF2 DecisionQuotient.Physics.ConstraintForcing.logic_time_not_sufficient_for_unique_law
CF3 DecisionQuotient.Physics.ConstraintForcing.laws_determined_implies_objective_determined
CF4 DecisionQuotient.Physics.ConstraintForcing.objective_not_determined_of_parameter_separation
CF5 DecisionQuotient.Physics.ConstraintForcing.forcedDecisionBits_pos_of_deadline
CF6 DecisionQuotient.Physics.ConstraintForcing.actionForced_of_deadline
CF7 DecisionQuotient.Physics.ConstraintForcing.nondegenerateBelief_of_deadline_and_uncertainty
CF8 DecisionQuotient.Physics.ConstraintForcing.forced_decision_implies_positive_landauer_cost
CF9 DecisionQuotient.Physics.ConstraintForcing.forced_decision_implies_positive_nv_work
CH1 DecisionQuotient.ClaimClosure.CH1
CH2 DecisionQuotient.ClaimClosure.CH2
CH3 DecisionQuotient.ClaimClosure.CH3

Lean handle entry (continued)

CH5 DecisionQuotient.ClaimClosure.CH5

CH6 DecisionQuotient.ClaimClosure.CH6

CIA1 ClaimClosure.ClassicalInfoAssumptions

COH1 dof_one_implies_coherent

COH2 dof_gt_one_incoherence_possible

COH3 determinate_truth_forces_ssot

CPL1 cost_ratio_eq_dof

CR1 DecisionQuotient.ConfigReduction.config_sufficiency_iff_behavior_preserving

CT1 DecisionQuotient.Physics.ClaimTransport.PhysicalEncoding

CT2 DecisionQuotient.Physics.ClaimTransport.physical_claim_lifts_from_core

CT3 DecisionQuotient.Physics.ClaimTransport.physical_claim_lifts_from_core_conditional

CT4 DecisionQuotient.Physics.ClaimTransport.physical_counterexample_yields_core_counterexample

CT5 DecisionQuotient.Physics.ClaimTransport.physical_counterexample_invalidates_core_rule

CT6 DecisionQuotient.Physics.ClaimTransport.no_physical_counterexample_of_core_theorem

CT7 DecisionQuotient.Physics.ClaimTransport.LawGapInstance

CT8 DecisionQuotient.Physics.ClaimTransport.lawGapEncoding

CT9 DecisionQuotient.Physics.ClaimTransport.lawGapPhysicalClaim

CT10 DecisionQuotient.Physics.ClaimTransport.law_gap_physical_claim_holds

CT11 DecisionQuotient.Physics.ClaimTransport.no_law_gap_counterexample

CT12 DecisionQuotient.Physics.ClaimTransport.physical_bridge_bundle

CV1 DecisionQuotient.Physics.Conversation.RecurrentCircuit

CV2 DecisionQuotient.Physics.Conversation.CoupledConversation

CV3 DecisionQuotient.Physics.Conversation.JointState

CV4 DecisionQuotient.Physics.Conversation.tick_uses_shared_node

CV5 DecisionQuotient.Physics.Conversation.tick_shared_is_merged_emissions

CV6 DecisionQuotient.Physics.Conversation.channel_projection_eq_iff_quantized_eq

CV7 DecisionQuotient.Physics.Conversation.clamp_projection_eq_iff_same_clamped_bit

CV10 DecisionQuotient.Physics.Conversation.explanationGap_add_explanationBits

CV11 DecisionQuotient.Physics.Conversation.toClaimReport

CV12 DecisionQuotient.Physics.Conversation.abstain_iff_no_answer

CV13 DecisionQuotient.Physics.Conversation.yes_no_iff_exact_claim

CV14 DecisionQuotient.Physics.Conversation.toReportSignal_completion_defined_of_budget

CV15 DecisionQuotient.Physics.Conversation.toReportSignal_signal_consistent_zero_certified

CV16 DecisionQuotient.Physics.Conversation.abstain_report_can_carry_explanation

CV17 DecisionQuotient.Physics.Conversation.clampDecisionEvent_iff_bitOps_pos

CV18 DecisionQuotient.Physics.Conversation.clamp_event_implies_positive_energy

CV19 DecisionQuotient.Physics.Conversation.BinaryAnswer

CV20 DecisionQuotient.Physics.Conversation.ConversationReport

CV21 DecisionQuotient.Physics.Conversation.explanationGap

CV26 DecisionQuotient.Physics.Conversation.toClaimReport_ne_epsilon

CV27 DecisionQuotient.Physics.Conversation.toReportSignal

CV28 DecisionQuotient.Physics.Conversation.toReportSignal_declares_bound

DC1 DecisionQuotient.StochasticSequential.static_stochastic_strict_separation

Lean handle entry (continued)
DC2 DecisionQuotient.StochasticSequential.stochastic_sequential_strict_separation
DC3 DecisionQuotient.StochasticSequential.complexity_dichotomy_hierarchy
DC4 DecisionQuotient.StochasticSequential.regime_hierarchy
DC5 DecisionQuotient.StochasticSequential.coNP_subset_PP
DC6 DecisionQuotient.StochasticSequential.PP_subset_PSPACE
DC7 DecisionQuotient.StochasticSequential.coNP_subset_PSPACE
DC8 DecisionQuotient.StochasticSequential.static_to_coNP
DC9 DecisionQuotient.StochasticSequential.stochastic_to_PP
DC10 DecisionQuotient.StochasticSequential.sequential_to_PSPACE
DC11 DecisionQuotient.StochasticSequential.ClaimClosure.claim_six_subcases
DC12 DecisionQuotient.StochasticSequential.ClaimClosure.claim_hierarchy
DC13 DecisionQuotient.StochasticSequential.ClaimClosure.claim_tractable_subcases_to_P
DC14 DecisionQuotient.StochasticSequential.stochastic_dichotomy
DC15 DecisionQuotient.StochasticSequential.above_threshold_hard
DC16 DecisionQuotient.StochasticSequential.StochasticAnchorSufficient
DC17 DecisionQuotient.StochasticSequential.StochasticAnchorSufficiencyCheck
DC18 DecisionQuotient.StochasticSequential.stochastic_anchor_check_iff
DC19 DecisionQuotient.StochasticSequential.stochastic_anchor_sufficient_of_stochastic_sufficient
DC20 DecisionQuotient.StochasticSequential.SequentialAnchorSufficient
DC21 DecisionQuotient.StochasticSequential.SequentialAnchorSufficiencyCheck
DC22 DecisionQuotient.StochasticSequential.sequential_anchor_check_iff
DC23 DecisionQuotient.StochasticSequential.sequential_anchor_sufficient_of_sequential_sufficient
DC24 DecisionQuotient.StochasticSequential.StochasticAnchorCheckInstance
DC25 DecisionQuotient.StochasticSequential.reduceMAJSAT_correct_anchor_strict
DC26 DecisionQuotient.StochasticSequential.reduceMAJSAT_to_stochastic_anchor_reduction
DC27 DecisionQuotient.StochasticSequential.SequentialAnchorCheckInstance
DC28 DecisionQuotient.StochasticSequential.reduceTQBF_correct_anchor
DC29 DecisionQuotient.StochasticSequential.reduceTQBF_to_sequential_anchor_reduction
DC30 DecisionQuotient.StochasticSequential.StatePotential
DC31 DecisionQuotient.StochasticSequential.utilityFromPotentialDrop_le_iff_nextPotential_ge
DC32 DecisionQuotient.StochasticSequential.utility_from_action_state_potential
DC33 DecisionQuotient.StochasticSequential.stochasticExpectedUtility_eq_neg_expectedActionPotential
DC34 DecisionQuotient.StochasticSequential.stochasticExpectedUtility_le_iff_expectedActionPotential_ge
DC35 DecisionQuotient.StochasticSequential.landauerEnergyFloor_nonneg
DC36 DecisionQuotient.StochasticSequential.landauerEnergyFloor_mono_bits
DC37 DecisionQuotient.StochasticSequential.thermodynamicCost_eq_landauerEnergyFloorRoom_states
DE1 DecisionQuotient.ClaimClosure.DE1
DE2 DecisionQuotient.ClaimClosure.DE2
DE3 DecisionQuotient.ClaimClosure.DE3
DE4 DecisionQuotient.ClaimClosure.DE4
DER1 all_derived_from_source
DER2 ClaimClosure.derivation_preserves_coherence_core
DES1 ClaimClosure.design_necessity

Lean handle entry (continued)
DG1 DecisionQuotient.Outside
DG2 DecisionQuotient.anchoredSlice
DG3 DecisionQuotient.anchoredSliceEquivOutside
DG4 DecisionQuotient.card_outside_eq_sub
DG5 DecisionQuotient.card_anchoredSlice
DG6 DecisionQuotient.card_anchoredSlice_eq_pow_sub
DG7 DecisionQuotient.card_anchoredSlice_eq_uniform
DG8 DecisionQuotient.anchoredSlice_mul_fixed_eq_full
DG9 DecisionQuotient.constantBoolDP
DG10 DecisionQuotient.firstCoordDP
DG11 DecisionQuotient.constantBoolDP_opt
DG12 DecisionQuotient.firstCoordDP_opt
DG13 DecisionQuotient.constantBoolDP_empty_sufficient
DG14 DecisionQuotient.firstCoordDP_empty_not_sufficient
DG15 DecisionQuotient.boolHypercube_node_count
DG16 DecisionQuotient.node_count_does_not_determine_edge_geometry
DG17 DecisionQuotient.DecisionProblem.edgeOnComplement
DG18 DecisionQuotient.DecisionProblem.edgeOnComplement_iff_not_sufficient
DP1 DecisionQuotient.DecisionProblem.minimalSufficient_iff_relevant
DP2 DecisionQuotient.DecisionProblem.relevantSet_is_minimal
DP3 DecisionQuotient.DecisionProblem.sufficient_implies_selectorSufficient
DP4 DecisionQuotient.ClaimClosure.DecisionProblem.epsOpt_zero_eq_opt
DP5 DecisionQuotient.ClaimClosure.DecisionProblem.sufficient_iff_zeroEpsilonSufficient
DP6 DecisionQuotient.ClaimClosure.DP6
DP7 DecisionQuotient.ClaimClosure.DP7
DP8 DecisionQuotient.ClaimClosure.DP8
DQ1 DecisionQuotient.ClaimClosure.DQ1
DQ2 DecisionQuotient.ClaimClosure.DQ2
DQ3 DecisionQuotient.ClaimClosure.DQ3
DQ4 DecisionQuotient.ClaimClosure.DQ4
DQ5 DecisionQuotient.ClaimClosure.DQ5
DQ6 DecisionQuotient.ClaimClosure.DQ6
DQ7 DecisionQuotient.ClaimClosure.DQ7
DQ8 DecisionQuotient.ClaimClosure.DQ8
DQ_16442f86 DQ.Sigma2PHardness.exactlyIdentifiesRelevant_iff_sufficient_and_subset_relevantFinset
DQ_5daf095c DQ.ClaimClosure.tractable_bounded_core
DQ_6925a675 DQ.ClaimClosure.tractable_tree_core
DQ_815ae965 DQ.ClaimClosure.process_bridge_failure_witness
DQ_8b434839 DQ.ClaimClosure.information_barrier_opt_oracle_core
DQ_8e70c0eb DQ.ClaimClosure.hard_family_all_coords_core
DQ_8ed12c6b DQ.ClaimClosure.horizonTwoWitness_immediate_empty_sufficient
DQ_926a3bda DQ.Sigma2PHardness.representationGap_eq_zero_iff
DQ_9460340e DQ.StochasticSequential.stochastic_sequential_strict_separation

Lean handle entry (continued)

DQ_951c2d27 DQ.Sigma2PHardness.min_representationGap_zero_iff_relevant_card

DQ_ab238b52 DQ.StochasticSequential.static_stochastic_strict_separation

DQ_bc3e53ee DQ.ClaimClosure.tractable_separable_core

DQ_e2dde061 DQ.ClaimClosure.horizon_gt_one_bridge_can_fail_on_sufficiency

DS1 DecisionQuotient.ClaimClosure.DS1

DS2 DecisionQuotient.ClaimClosure.DS2

DS3 DecisionQuotient.ClaimClosure.DS3

DS4 DecisionQuotient.ClaimClosure.DS4

DS5 DecisionQuotient.ClaimClosure.DS5

DS6 DecisionQuotient.ClaimClosure.DS6

DT1 DecisionQuotient.Physics.DecisionTime.TimedState

DT2 DecisionQuotient.Physics.DecisionTime.DecisionProcess

DT3 DecisionQuotient.Physics.DecisionTime.tick

DT4 DecisionQuotient.Physics.DecisionTime.DecisionEvent

DT5 DecisionQuotient.Physics.DecisionTime.TimeUnitStep

DT6 DecisionQuotient.Physics.DecisionTime.time_is_discrete

DT7 DecisionQuotient.Physics.DecisionTime.time_coordinate_falsifiable

DT8 DecisionQuotient.Physics.DecisionTime.tick_increments_time

DT9 DecisionQuotient.Physics.DecisionTime.tick_decision_witness

DT10 DecisionQuotient.Physics.DecisionTime.tick_is_decision_event

DT11 DecisionQuotient.Physics.DecisionTime.decision_event_implies_time_unit

DT12 DecisionQuotient.Physics.DecisionTime.decision_taking_place_is_unit_of_time

DT13 DecisionQuotient.Physics.DecisionTime.decision_event_iff_eq_tick

DT14 DecisionQuotient.Physics.DecisionTime.run

DT15 DecisionQuotient.Physics.DecisionTime.run_time_exact

DT16 DecisionQuotient.Physics.DecisionTime.run_elapsed_time_eq_ticks

DT17 DecisionQuotient.Physics.DecisionTime.decisionTrace

DT18 DecisionQuotient.Physics.DecisionTime.decisionTrace_length_eq_ticks

DT19 DecisionQuotient.Physics.DecisionTime.decision_count_equals_elapsed_time

DT20 DecisionQuotient.Physics.DecisionTime.SubstrateKind

DT21 DecisionQuotient.Physics.DecisionTime.SubstrateModel

DT22 DecisionQuotient.Physics.DecisionTime.substrate_step_realizes_decision_event

DT23 DecisionQuotient.Physics.DecisionTime.substrate_step_is_time_unit

DT24 DecisionQuotient.Physics.DecisionTime.time_unit_law_substrate_invariant

EI1 DecisionQuotient.ThermodynamicLift.energy_ge_kbt_nat_entropy

EI2 DecisionQuotient.ThermodynamicLift.energy_ground_state_tracks_entropy

EI4 DecisionQuotient.ThermodynamicLift.landauerJoulesPerBit_pos

EI5 DecisionQuotient.ThermodynamicLift.neukart_vinokur_duality

ENT1 Entropy.ClassicalEntropyAssumptions

EP1 DecisionQuotient.Physics.LocalityPhysics.landauer_principle

EP2 DecisionQuotient.Physics.LocalityPhysics.finite_regional_energy

EP3 DecisionQuotient.Physics.LocalityPhysics.finite_signal_speed

EP4 DecisionQuotient.Physics.LocalityPhysics.nontrivial_physics

Lean handle entry (continued)
FI3 DecisionQuotient.FunctionalInformation.functional_information_from_counting
FI6 DecisionQuotient.FunctionalInformation.functional_information_from_thermodynamics
FI7 DecisionQuotient.FunctionalInformation.first_principles_thermo_coincide
FLP1 ClaimClosure.static_flp_core
FN7 DecisionQuotient.BayesOptimalityProof.KL_nonneg
FN8 DecisionQuotient.BayesOptimalityProof.entropy_le_crossEntropy
FN12 DecisionQuotient.BayesOptimalityProof.crossEntropy_eq_entropy_add_KL
FN14 DecisionQuotient.BayesOptimalityProof.bayes_is_optimal
FN15 DecisionQuotient.BayesOptimalityProof.KL_eq_sum_klFun
FN16 DecisionQuotient.BayesOptimalityProof.KL_eq_zero_iff_eq
FP1 DecisionQuotient.Physics.LocalityPhysics.trivial_states_all_equal
FP2 DecisionQuotient.Physics.LocalityPhysics.equal_states_constant_function
FP3 DecisionQuotient.Physics.LocalityPhysics.constant_function_singleton_image
FP4 DecisionQuotient.Physics.LocalityPhysics.singleton_image_zero_entropy
FP5 DecisionQuotient.Physics.LocalityPhysics.zero_entropy_no_information
FP6 DecisionQuotient.Physics.LocalityPhysics.triviality_implies_no_information
FP7 DecisionQuotient.Physics.LocalityPhysics.information_requires_nontriviality
FP8 DecisionQuotient.Physics.LocalityPhysics.atypical_states_rare
FP9 DecisionQuotient.Physics.LocalityPhysics.random_misses_target
FP10 DecisionQuotient.Physics.LocalityPhysics.errors_accumulate
FP11 DecisionQuotient.Physics.LocalityPhysics.wrong_paths_dominate
FP12 DecisionQuotient.Physics.LocalityPhysics.second_law_from_counting
FP13 DecisionQuotient.Physics.LocalityPhysics.verification_is_information
FP14 DecisionQuotient.Physics.LocalityPhysics.entropy_is_information
FP15 DecisionQuotient.Physics.LocalityPhysics.landauer_structure
FPT1 DecisionQuotient.Physics.LocalityPhysics.Timeline
FPT2 DecisionQuotient.Physics.LocalityPhysics.FPT2_function_deterministic
FPT3 DecisionQuotient.Physics.LocalityPhysics.FPT3_outputs_differ_inputs_differ
FPT4 DecisionQuotient.Physics.LocalityPhysics.FPT4_step_requires_distinct_moments
FPT5 DecisionQuotient.Physics.LocalityPhysics.FPT5_distinct_moments_positive_duration
FPT6 DecisionQuotient.Physics.LocalityPhysics.FPT6_step_takes_positive_time
FPT7 DecisionQuotient.Physics.LocalityPhysics.FPT7_no_instantaneous_steps
FPT8 DecisionQuotient.Physics.LocalityPhysics.FPT8_propagation_takes_time
FPT9 DecisionQuotient.Physics.LocalityPhysics.FPT9_speed_bounded_by_positive_time
FPT10 DecisionQuotient.Physics.LocalityPhysics.FPT10_ec3_is_logical
FS1 DecisionQuotient.Statistics.sum_fisherScore_eq_srank
FS2 DecisionQuotient.Statistics.fisherMatrix_rank_eq_srank
FS3 DecisionQuotient.Statistics.fisherMatrix_trace_eq_srank
FS4 DecisionQuotient.Statistics.fisherScore_relevant
FS5 DecisionQuotient.Statistics.fisherScore_irrelevant
GE1 DecisionQuotient.ClaimClosure.GE1
GE2 DecisionQuotient.ClaimClosure.GE2
GE3 DecisionQuotient.ClaimClosure.GE3

Lean handle entry (continued)
GE4 DecisionQuotient.ClaimClosure.GE4
GE5 DecisionQuotient.ClaimClosure.GE5
GE6 DecisionQuotient.ClaimClosure.GE6
GE7 DecisionQuotient.ClaimClosure.GE7
GE8 DecisionQuotient.ClaimClosure.GE8
GE9 DecisionQuotient.ClaimClosure.GE9
GEN1 generated_file_is_second_encoding
GN1 DecisionQuotient.LogicGraph.isCycle
GN2 DecisionQuotient.LogicGraph.cycleWitnessBits_pos
GN3 DecisionQuotient.LogicGraph.pathProb_nonneg
GN4 DecisionQuotient.LogicGraph.pathSurprisal_nonneg_of_positive_mass
GN5 DecisionQuotient.LogicGraph.nontrivialityScore_unknown
GN6 DecisionQuotient.LogicGraph.observerEntropy_nonneg
GN7 DecisionQuotient.LogicGraph.dqFromEntropy_in_unit_interval
GN8 DecisionQuotient.LogicGraph.path_belief_forced_under_uncertainty
GN9 DecisionQuotient.LogicGraph.bayes_update_exists_for_observer_paths
GN10 DecisionQuotient.LogicGraph.cycle_witness_implies_positive_landauer
GN11 DecisionQuotient.LogicGraph.cycle_witness_implies_positive_nv_work
GN12 DecisionQuotient.LogicGraph.dna_erasure_implies_positive_landauer
GN13 DecisionQuotient.LogicGraph.dna_room_temp_environmental_stability
HD1 DecisionQuotient.HardnessDistribution.centralization_dominance_bundle
HD2 DecisionQuotient.HardnessDistribution.centralization_step_saves_n_minus_one
HD3 DecisionQuotient.HardnessDistribution.centralized_higher_leverage
HD4 DecisionQuotient.HardnessDistribution.complete_model_dominates_after_threshold
HD5 DecisionQuotient.HardnessDistribution.gap_conservation_card
HD6 DecisionQuotient.HardnessDistribution.generalizedTotal_with_saturation_eventually_constant
HD7 DecisionQuotient.HardnessDistribution.generalized_dominance_can_fail_without_right_boundedness
HD8 DecisionQuotient.HardnessDistribution.generalized_dominance_can_fail_without_wrong_growth
HD9 DecisionQuotient.HardnessDistribution.generalized_right_dominates_wrong_of_bounded_vs_identity_lower
HD10 DecisionQuotient.HardnessDistribution.generalized_right_eventually_dominates_wrong
HD11 DecisionQuotient.HardnessDistribution.hardnessEfficiency_eq_central_share
HD12 DecisionQuotient.HardnessDistribution.isRightHardness
HD13 DecisionQuotient.HardnessDistribution.isWrongHardness
HD14 DecisionQuotient.HardnessDistribution.linear_lt_exponential_plus_constant_eventually
HD15 DecisionQuotient.HardnessDistribution.native_dominates_manual
HD16 DecisionQuotient.HardnessDistribution.no_positive_slope_linear_represents_saturating
HD17 DecisionQuotient.HardnessDistribution.requiredWork
HD18 DecisionQuotient.HardnessDistribution.requiredWork_eq_affine_in_sites
HD19 DecisionQuotient.HardnessDistribution.right_dominates_wrong
HD20 DecisionQuotient.HardnessDistribution.saturatingSiteCost_eventually_constant
HD21 DecisionQuotient.HardnessDistribution.simplicityTax_grows
HD22 DecisionQuotient.HardnessDistribution.hardnessLowerBound
HD23 DecisionQuotient.HardnessDistribution.hardness_is_irreducible_required_work

Lean handle entry (continued)
HD24 DecisionQuotient.HardnessDistribution.totalDOF_eventually_constant_iff_zero_distributed
HD25 DecisionQuotient.HardnessDistribution.totalDOF_ge_intrinsic
HD26 DecisionQuotient.HardnessDistribution.totalExternalWork_eq_n_mul_gapCard
HD27 DecisionQuotient.HardnessDistribution.workGrowthDegree
HD28 DecisionQuotient.HardnessDistribution.workGrowthDegree_zero_iff_eventually_constant
HS1 DecisionQuotient.Physics.HeisenbergStrong.NoisyPhysicalEncoding
HS2 DecisionQuotient.Physics.HeisenbergStrong.HeisenbergStrongBinding
HS3 DecisionQuotient.Physics.HeisenbergStrong.strong_binding_implies_core_nontrivial
HS4 DecisionQuotient.Physics.HeisenbergStrong.strong_binding_yields_core_encoding_witness
HS5 DecisionQuotient.Physics.HeisenbergStrong.strong_binding_implies_physical_nontrivial_opt_assumption
HS6 DecisionQuotient.Physics.HeisenbergStrong.strong_binding_implies_nontrivial_opt_via_uncertainty
IA1 DecisionQuotient.ClaimClosure.IA1
IA2 DecisionQuotient.ClaimClosure.IA2
IA3 DecisionQuotient.ClaimClosure.IA3
IA4 DecisionQuotient.ClaimClosure.IA4
IA5 DecisionQuotient.ClaimClosure.IA5
IA6 DecisionQuotient.ClaimClosure.IA6
IA7 DecisionQuotient.ClaimClosure.IA7
IA8 DecisionQuotient.ClaimClosure.IA8
IA9 DecisionQuotient.ClaimClosure.IA9
IA10 DecisionQuotient.ClaimClosure.IA10
IA11 DecisionQuotient.ClaimClosure.IA11
IA12 DecisionQuotient.ClaimClosure.IA12
IA13 DecisionQuotient.ClaimClosure.IA13
IC1 DecisionQuotient.IntegrityCompetence.CertaintyInflation
IC2 DecisionQuotient.IntegrityCompetence.CompletionFractionDefined
IC3 DecisionQuotient.IntegrityCompetence.EvidenceForReport
IC4 DecisionQuotient.IntegrityCompetence.ExactCertaintyInflation
IC5 DecisionQuotient.IntegrityCompetence.Percent
IC6 DecisionQuotient.IntegrityCompetence.RLFFWeights
IC7 DecisionQuotient.IntegrityCompetence.ReportSignal
IC8 DecisionQuotient.IntegrityCompetence.ReportBitModel
IC9 DecisionQuotient.IntegrityCompetence.SignalConsistent
IC10 DecisionQuotient.IntegrityCompetence.admissible_irrational_strictly_more_than_rational
IC11 DecisionQuotient.IntegrityCompetence.admissible_matrix_counts
IC12 DecisionQuotient.IntegrityCompetence.abstain_signal_exists_with_guess_self
IC13 DecisionQuotient.IntegrityCompetence.certaintyInflation_iff_not_admissible
IC14 DecisionQuotient.IntegrityCompetence.certificationOverheadBits
IC15 DecisionQuotient.IntegrityCompetence.certificationOverheadBits_of_evidence
IC16 DecisionQuotient.IntegrityCompetence.certificationOverheadBits_of_no_evidence
IC17 DecisionQuotient.IntegrityCompetence.certifiedTotalBits
IC18 DecisionQuotient.IntegrityCompetence.certifiedTotalBits_ge_raw
IC19 DecisionQuotient.IntegrityCompetence.certifiedTotalBits_gt_raw_of_evidence

Lean handle entry (continued)
IC20 DecisionQuotient.IntegrityCompetence.certifiedTotalBits_of_evidence
IC21 DecisionQuotient.IntegrityCompetence.certifiedTotalBits_of_no_evidence
IC22 DecisionQuotient.IntegrityCompetence.claim_admissible_of_evidence
IC23 DecisionQuotient.IntegrityCompetence.competence_implies_integrity
IC24 DecisionQuotient.IntegrityCompetence.completion_fraction_defined_of_declared_bound
IC25 DecisionQuotient.IntegrityCompetence.epsilon_competence_implies_integrity
IC26 DecisionQuotient.IntegrityCompetence.evidence_nonempty_iff_claim_admissible
IC27 DecisionQuotient.IntegrityCompetence.evidence_of_claim_admissible
IC28 DecisionQuotient.IntegrityCompetence.exact_claim_admissible_iff_exact_evidence_nonempty
IC29 DecisionQuotient.IntegrityCompetence.exact_claim_requires_evidence
IC30 DecisionQuotient.IntegrityCompetence.exactCertaintyInflation_iff_no_exact_competence
IC31 DecisionQuotient.IntegrityCompetence.exact_raw_only_of_no_exact_admissible
IC32 DecisionQuotient.IntegrityCompetence.integrity_forces_abstention
IC33 DecisionQuotient.IntegrityCompetence.integrity_not_competent_of_nonempty_scope
IC34 DecisionQuotient.IntegrityCompetence.integrity_resource_bound
IC35 DecisionQuotient.IntegrityCompetence.no_completion_fraction_without_declared_bound
IC36 DecisionQuotient.IntegrityCompetence.overModelVerdict_rational_iff
IC37 DecisionQuotient.IntegrityCompetence.percentZero
IC38 DecisionQuotient.IntegrityCompetence.rlffBaseReward
IC39 DecisionQuotient.IntegrityCompetence.rlffReward
IC40 DecisionQuotient.IntegrityCompetence.rlff_abstain_strictly_prefers_no_certificates
IC41 DecisionQuotient.IntegrityCompetence.rlff_maximizer_has_evidence
IC42 DecisionQuotient.IntegrityCompetence.rlff_maximizer_is_admissible
IC43 DecisionQuotient.IntegrityCompetence.self_reflected_confidence_not_certification
IC44 DecisionQuotient.IntegrityCompetence.signal_certified_positive_implies_admissible
IC45 DecisionQuotient.IntegrityCompetence.signal_consistent_of_claim_admissible
IC46 DecisionQuotient.IntegrityCompetence.signal_no_evidence_forces_zero_certified
IC47 DecisionQuotient.IntegrityCompetence.signal_exact_no_competence_forces_zero_certified
IC48 DecisionQuotient.IntegrityCompetence.steps_run_scalar_always_defined
IC49 DecisionQuotient.IntegrityCompetence.steps_run_scalar_falsifiable
IC50 DecisionQuotient.IntegrityCompetence.zero_epsilon_competence_iff_exact
IE1 DecisionQuotient.ClaimClosure.IE1
IE2 DecisionQuotient.ClaimClosure.IE2
IE3 DecisionQuotient.ClaimClosure.IE3
IE4 DecisionQuotient.ClaimClosure.IE4
IE5 DecisionQuotient.ClaimClosure.IE5
IE6 DecisionQuotient.ClaimClosure.IE6
IE7 DecisionQuotient.ClaimClosure.IE7
IE8 DecisionQuotient.ClaimClosure.IE8
IE9 DecisionQuotient.ClaimClosure.IE9
IE10 DecisionQuotient.ClaimClosure.IE10
IE11 DecisionQuotient.ClaimClosure.IE11
IE12 DecisionQuotient.ClaimClosure.IE12

Lean handle entry (continued)
IE13 DecisionQuotient.ClaimClosure.IE13
IE14 DecisionQuotient.ClaimClosure.IE14
IE15 DecisionQuotient.ClaimClosure.IE15
IE16 DecisionQuotient.ClaimClosure.IE16
IE17 DecisionQuotient.ClaimClosure.IE17
IN1 DecisionQuotient.Physics.Instantiation.Geometry
IN2 DecisionQuotient.Physics.Instantiation.Dynamics
IN3 DecisionQuotient.Physics.Instantiation.Circuit
IN4 DecisionQuotient.Physics.Instantiation.geometry_plus_dynamics_is_circuit
IN5 DecisionQuotient.Physics.Instantiation.DecisionInterpretation
IN6 DecisionQuotient.Physics.Instantiation.DecisionCircuit
IN7 DecisionQuotient.Physics.Instantiation.Molecule
IN8 DecisionQuotient.Physics.Instantiation.Reaction
IN9 DecisionQuotient.Physics.Instantiation.ReactionOutcome
IN10 DecisionQuotient.Physics.Instantiation.MoleculeGeometry
IN11 DecisionQuotient.Physics.Instantiation.MoleculeDynamics
IN12 DecisionQuotient.Physics.Instantiation.MoleculeCircuit
IN13 DecisionQuotient.Physics.Instantiation.MoleculeAsCircuit
IN14 DecisionQuotient.Physics.Instantiation.MoleculeAsDecisionCircuit
IN15 DecisionQuotient.Physics.Instantiation.molecule_decision_preserves_geometry
IN16 DecisionQuotient.Physics.Instantiation.molecule_decision_preserves_dynamics
IN17 DecisionQuotient.Physics.Instantiation.asDecisionCircuit
IN18 DecisionQuotient.Physics.Instantiation.asDecisionCircuit_preserves_circuit
IN19 DecisionQuotient.Physics.Instantiation.Configuration
IN20 DecisionQuotient.Physics.Instantiation.EnergyLandscape
IN21 DecisionQuotient.Physics.Instantiation.k_Boltzmann
IN22 DecisionQuotient.Physics.Instantiation.LandauerBound
IN23 DecisionQuotient.Physics.Instantiation.law_objective_schema
IN24 DecisionQuotient.Physics.Instantiation.law_opt_eq_feasible_of_gap
IN25 DecisionQuotient.Physics.Instantiation.law_opt_singleton_of_deterministic
IND1 both_requirements_independent
IND2 both_requirements_independent'
INS1 Inconsistency.ssot_is_unique_optimum
INS2 Inconsistency.ssot_required
INS3 Inconsistency.ssot_unique_satisfier
INS4 Inconsistency.resolution_requires_external_choice
IP1 DecisionQuotient.Physics.LocalityPhysics.ec1_can_be_true
IP2 DecisionQuotient.Physics.LocalityPhysics.ec1_independent_of_ec2_ec3
IP3 DecisionQuotient.Physics.LocalityPhysics.ec2_can_be_true
IP4 DecisionQuotient.Physics.LocalityPhysics.ec2_independent_of_ec1_ec3
IP5 DecisionQuotient.Physics.LocalityPhysics.ec3_can_be_true
IP6 DecisionQuotient.Physics.LocalityPhysics.ec3_independent_of_ec1_ec2
IP7 DecisionQuotient.Physics.LocalityPhysics.empirical_claims_mutually_independent

Lean handle entry (continued)
IT1 DecisionQuotient.DecisionProblem.numOptClasses
IT3 DecisionQuotient.quotientEntropy_le_srank_binary
IT4 DecisionQuotient.numOptClasses_le_pow_srank_binary
IT5 DecisionQuotient.nontrivial_bounds_binary
IT6 DecisionQuotient.nontrivial_implies_srank_pos
IV1 DecisionQuotient.InteriorVerification.GoalClass
IV2 DecisionQuotient.InteriorVerification.InteriorDominanceVerifiable
IV3 DecisionQuotient.InteriorVerification.TautologicalSetIdentifiable
IV4 DecisionQuotient.InteriorVerification.agreeOnSet
IV5 DecisionQuotient.InteriorVerification.interiorParetoDominates
IV6 DecisionQuotient.InteriorVerification.interior_certificate_implies_non_rejection
IV7 DecisionQuotient.InteriorVerification.interior_dominance_implies_universal_non_rejection
IV8 DecisionQuotient.InteriorVerification.interior_dominance_not_full_sufficiency
IV9 DecisionQuotient.InteriorVerification.interior_verification_tractable_certificate
IV10 DecisionQuotient.InteriorVerification.not_sufficientOnSet_of_counterexample
IV11 DecisionQuotient.InteriorVerification.singleton_coordinate_interior_certificate
LNG1 ClaimClosure.language_realizability_criterion
LP1 DecisionQuotient.Physics.LocalityPhysics.SpacetimePoint
LP2 DecisionQuotient.Physics.LocalityPhysics.lightCone
LP3 DecisionQuotient.Physics.LocalityPhysics.causalPast
LP4 DecisionQuotient.Physics.LocalityPhysics.self_in_lightCone
LP5 DecisionQuotient.Physics.LocalityPhysics.self_in_causalPast
LP6 DecisionQuotient.Physics.LocalityPhysics.LocalRegion
LP7 DecisionQuotient.Physics.LocalityPhysics.canObserve
LP8 DecisionQuotient.Physics.LocalityPhysics.spacelikeSeparated
LP9 DecisionQuotient.Physics.LocalityPhysics.spacelike_disjoint_observation
LP11 DecisionQuotient.Physics.LocalityPhysics.LocalConfiguration
LP12 DecisionQuotient.Physics.LocalityPhysics.isLocallyValid
LP13 DecisionQuotient.Physics.LocalityPhysics.MergeResult
LP14 DecisionQuotient.Physics.LocalityPhysics.boardMerge
LP15 DecisionQuotient.Physics.LocalityPhysics.independent_configs_can_disagree
LP16 DecisionQuotient.Physics.LocalityPhysics.merge_compatible_iff
LP17 DecisionQuotient.Physics.LocalityPhysics.merge_contradiction_iff
LP18 DecisionQuotient.Physics.LocalityPhysics.locality_implies_possible_contradiction
LP19 DecisionQuotient.Physics.LocalityPhysics.Superposition
LP20 DecisionQuotient.Physics.LocalityPhysics.superposition_can_contain_contradictions
LP21 DecisionQuotient.Physics.LocalityPhysics.superposition_requires_separation
LP22 DecisionQuotient.Physics.LocalityPhysics.bell_separation_is_real
LP23 DecisionQuotient.Physics.LocalityPhysics.measurement_is_merge_contradiction
LP24 DecisionQuotient.Physics.LocalityPhysics.entanglement_is_shared_origin
LP31 DecisionQuotient.Physics.LocalityPhysics.complete_knowledge_requires_all_queries
LP32 DecisionQuotient.Physics.LocalityPhysics.finite_energy_constraint
LP33 DecisionQuotient.Physics.LocalityPhysics.self_knowledge_impossible_if_insufficient_energy

Lean handle entry (continued)
LP34 DecisionQuotient.Physics.LocalityPhysics.bounded_energy_forces_locality
LP35 DecisionQuotient.Physics.LocalityPhysics.locality_implies_independent_regions
LP36 DecisionQuotient.Physics.LocalityPhysics.independent_regions_imply_possible_contradiction
LP38 DecisionQuotient.Physics.LocalityPhysics.pne_np_necessary_for_physics
LP39 DecisionQuotient.Physics.LocalityPhysics.matter_exists_because_pne_np
LP40 DecisionQuotient.Physics.LocalityPhysics.physics_is_the_game
LP41 DecisionQuotient.Physics.LocalityPhysics.without_positive_query_cost_no_bound
LP42 DecisionQuotient.Physics.LocalityPhysics.without_nontrivial_states_no_disagreement
LP43 DecisionQuotient.Physics.LocalityPhysics.without_separation_no_independence
LP44 DecisionQuotient.Physics.LocalityPhysics.without_finite_capacity_no_gap
LP45 DecisionQuotient.Physics.LocalityPhysics.all_premises_used
LP46 DecisionQuotient.Physics.LocalityPhysics.premises_necessary_and_sufficient
LP50 DecisionQuotient.Physics.LocalityPhysics.shannon_value_is_intractability
LP51 DecisionQuotient.Physics.LocalityPhysics.economic_value_requires_scarcity
LP52 DecisionQuotient.Physics.LocalityPhysics.thermodynamic_value_requires_gradient
LP53 DecisionQuotient.Physics.LocalityPhysics.voi_requires_uncertainty
LP54 DecisionQuotient.Physics.LocalityPhysics.physics_requires_intractability
LP55 DecisionQuotient.Physics.LocalityPhysics.value_is_intractability
LP56 DecisionQuotient.Physics.LocalityPhysics.observers_value_the_intractable
LP57 DecisionQuotient.Physics.LocalityPhysics.finite_steps_finite_coverage
LP58 DecisionQuotient.Physics.LocalityPhysics.counting_gap
LP59 DecisionQuotient.Physics.LocalityPhysics.time_is_counting
LP60 DecisionQuotient.Physics.LocalityPhysics.gap_equivalence
LP61 DecisionQuotient.Physics.LocalityPhysics.light_cone_is_time_gap
MI1 DecisionQuotient.ClaimClosure.MI1
MI2 DecisionQuotient.ClaimClosure.MI2
MI3 DecisionQuotient.ClaimClosure.MI3
MI4 DecisionQuotient.ClaimClosure.MI4
MI5 DecisionQuotient.ClaimClosure.MI5
MN1 DecisionQuotient.Physics.MeasureNecessity.quantitative_claim_has_measure
MN2 DecisionQuotient.Physics.MeasureNecessity.stochastic_claim_has_probability_measure
MN3 DecisionQuotient.Physics.MeasureNecessity.stochastic_claim_has_measure
MN4 DecisionQuotient.Physics.MeasureNecessity.count_univ_bool
MN5 DecisionQuotient.Physics.MeasureNecessity.counting_measure_not_probability_on_bool
MN6 DecisionQuotient.Physics.MeasureNecessity.deterministic_dirac_is_probability
MN7 DecisionQuotient.Physics.MeasureNecessity.quantitative_value_depends_on_measure
MN8 DecisionQuotient.Physics.MeasureNecessity.deterministic_models_still_measure_based
MN9 DecisionQuotient.Physics.MeasureNecessity.measure_does_not_imply_probability
MN10 DecisionQuotient.Physics.MeasureNecessity.quantitative_measure_is_logical_prerequisite
MN11 DecisionQuotient.Physics.MeasureNecessity.stochastic_probability_is_logical_prerequisite
OR1 DecisionQuotient.Physics.ObserverRelativeState.ObserverClass
OR2 DecisionQuotient.Physics.ObserverRelativeState.obsEquiv
OR3 DecisionQuotient.Physics.ObserverRelativeState.EffectiveStateSpace

Lean handle entry (continued)
OR4 DecisionQuotient.Physics.ObserverRelativeState.project_eq_iff
OR5 DecisionQuotient.Physics.ObserverRelativeState.observer_relative_equivalence_witness
OR6 DecisionQuotient.Physics.ObserverRelativeState.PhysicalObserverClass
OR7 DecisionQuotient.Physics.ObserverRelativeState.PhysicalStateSpace
OR8 DecisionQuotient.Physics.ObserverRelativeState.physical_state_space_has_instance_witness
OR9 DecisionQuotient.Physics.ObserverRelativeState.physical_observer_relative_effective_space
ORA1 oracle_arbitrary
PA1 DecisionQuotient.Physics.AnchorChecks.obsEquiv_all_of_effective_subsingleton
PA2 DecisionQuotient.Physics.AnchorChecks.stochasticAnchorSufficient_iff_exists_anchor_singleton
PA3 DecisionQuotient.Physics.AnchorChecks.stochastic_anchor_check_iff_exists_anchor_singleton
PA4 DecisionQuotient.Physics.AnchorChecks.stochastic_sufficient_of_observer_collapse_and_seed
PA5 DecisionQuotient.Physics.AnchorChecks.stochastic_anchor_check_of_observer_collapse_and_seed
PA6 DecisionQuotient.Physics.AnchorChecks.sequential_sufficient_of_observer_collapse
PA7 DecisionQuotient.Physics.AnchorChecks.sequential_anchor_check_of_observer_collapse
PA8 DecisionQuotient.Physics.AnchorChecks.physical_observer_collapse_implies_obsEquiv_all
PA9 DecisionQuotient.Physics.AnchorChecks.physical_stochastic_anchor_check_of_observer_collapse_and_seed
PBC1 DecisionQuotient.PhysicalBudgetCrossover.CrossoverAt
PBC2 DecisionQuotient.PhysicalBudgetCrossover.SuccinctInfeasible
PBC3 DecisionQuotient.PhysicalBudgetCrossover.SuccinctUnbounded
PBC4 DecisionQuotient.PhysicalBudgetCrossover.explicit_infeasible_succinct_feasible_of_crossover
PBC5 DecisionQuotient.PhysicalBudgetCrossover.exists_least_crossover_point
PBC6 DecisionQuotient.PhysicalBudgetCrossover.has_crossover_of_bounded_succinct_unbounded_explicit
PBC7 DecisionQuotient.PhysicalBudgetCrossover.explicit_eventual_infeasibility_of_monotone_and_witness
PBC8 DecisionQuotient.PhysicalBudgetCrossover.crossover_eventually_of_eventual_split
PBC9 DecisionQuotient.PhysicalBudgetCrossover.payoff_threshold_explicit_vs_succinct
PBC10 DecisionQuotient.PhysicalBudgetCrossover.no_universal_survivor_without_succinct_bound
PBC11 DecisionQuotient.PhysicalBudgetCrossover.policy_closure_at_divergence
PBC12 DecisionQuotient.PhysicalBudgetCrossover.policy_closure_beyond_divergence
PH1 PhysicalComplexity.k_Boltzmann
PH2 PhysicalComplexity.PhysicalComputer
PH3 PhysicalComplexity.bit_energy_cost
PH4 PhysicalComplexity.Landauer_bound
PH5 PhysicalComplexity.InstanceSize
PH6 PhysicalComplexity.ComputationalRequirement
PH7 PhysicalComplexity.coNP_requirement
PH8 PhysicalComplexity.coNP_physically_impossible
PH9 PhysicalComplexity.coNP_not_in_P_physically
PH10 PhysicalComplexity.sufficiency_physically_impossible
PH11 DecisionQuotient.PhysicalComplexity.PhysicalCollapseAtRequirement
PH12 DecisionQuotient.PhysicalComplexity.no_physical_collapse_at_requirement
PH13 DecisionQuotient.PhysicalComplexity.canonical_physical_collapse_impossible
PH14 DecisionQuotient.PhysicalComplexity.p_eq_np_physically_impossible_of_collapse_map
PH15 DecisionQuotient.PhysicalComplexity.p_eq_np_physically_impossible_canonical

Lean handle entry (continued)
PH16 DecisionQuotient.PhysicalComplexity.P_eq_NP_via_SAT
PH17 DecisionQuotient.PhysicalComplexity.SAT3ReductionBridge
PH18 DecisionQuotient.PhysicalComplexity.sat_reduction_transfers_energy_lower_bound
PH19 DecisionQuotient.PhysicalComplexity.physical_collapse_of_polytime_sat_realization
PH20 DecisionQuotient.PhysicalComplexity.p_eq_np_physically_impossible_via_sat_bridge
PH21 DecisionQuotient.PhysicalComplexity.SAT3HardFamily
PH22 DecisionQuotient.PhysicalComplexity.p_eq_np_physically_impossible_via_sat_hard_family
PH23 DecisionQuotient.PhysicalComplexity.collapse_possible_without_positive_bit_cost
PH24 DecisionQuotient.PhysicalComplexity.collapse_possible_without_exponential_lower_bound
PH25 DecisionQuotient.PhysicalComplexity.no_go_transfer_requires_collapse_map
PH26 DecisionQuotient.PhysicalComplexity.no_collapse_of_bounded_budget_pos_cost_exp_lb
PH27 DecisionQuotient.PhysicalComplexity.collapse_implies_assumption_failure_disjunction
PH28 DecisionQuotient.PhysicalComplexity.deterministic_no_physical_collapse
PH29 DecisionQuotient.PhysicalComplexity.probabilistic_no_physical_collapse
PH30 DecisionQuotient.PhysicalComplexity.sequential_no_physical_collapse
PH31 DecisionQuotient.PhysicalComplexity.collapse_possible_with_unbounded_budget_profile
PH32 DecisionQuotient.PhysicalComplexity.exp_budget_profile_unbounded
PH33 DecisionQuotient.PhysicalComplexity.finite_budget_assumption_is_necessary
PH34 DecisionQuotient.PhysicalComplexity.CoherentDQRejectionAtRequirement
PH35 DecisionQuotient.PhysicalComplexity.coherent_dq_rejection_impossible_at_requirement
PH36 DecisionQuotient.PhysicalComplexity.coherent_dq_rejection_impossible_canonical
PI1 DecisionQuotient.Physics.PhysicalIncompleteness.UniverseModel
PI2 DecisionQuotient.Physics.PhysicalIncompleteness.PhysicallyInstantiated
PI3 DecisionQuotient.Physics.PhysicalIncompleteness.no_surjective_instantiation_of_card_gap
PI4 DecisionQuotient.Physics.PhysicalIncompleteness.physical_incompleteness_of_card_gap
PI5 DecisionQuotient.Physics.PhysicalIncompleteness.physical_incompleteness_of_bounds
PI6 DecisionQuotient.Physics.PhysicalIncompleteness.under_resolution_implies_collision
PI7 DecisionQuotient.Physics.PhysicalIncompleteness.under_resolution_implies_decision_collision
PS1 DecisionQuotient.Physics.ClaimTransport.PhysicalStateSemantics
PS2 DecisionQuotient.Physics.ClaimTransport.physical_state_has_witness
PS3 DecisionQuotient.Physics.ClaimTransport.physical_state_claim_of_instance_claim
PS4 DecisionQuotient.Physics.ClaimTransport.physical_state_claim_of_universal_core
PYH1 Python.python_has_hooks
PYI1 Python.python_has_introspection
QT1 DecisionQuotient.DecisionProblem.quotient_is_coarsest
QT2 DecisionQuotient.DecisionProblem.quotientMap_preservesOpt
QT3 DecisionQuotient.DecisionProblem.quotient_represents_opt_equiv
QT4 DecisionQuotient.DecisionProblem.factors_implies_respects
RAT1 ClaimClosure.rate_incoherence_step
RD1 DecisionQuotient.Information.shannonEntropy_nonneg
RD2 DecisionQuotient.Information.rate_zero_distortion
RD3 DecisionQuotient.Information.rate_monotone
RED1 ClaimClosure.redundancy_incoherence_equiv

Lean handle entry (continued)
REG1 ClaimClosure.operating_regimes_partition
REG2 ClaimClosure.pareto_optimality_p1
REG3 ClaimClosure.no_tradeoff_at_p1
REG4 ClaimClosure.amortized_complexity_core
REQ1 structural_facts_fixed_at_definition
REQ2 definition_hooks_necessary
REQ3 introspection_necessary_for_verification
RS1 DecisionQuotient.Information.equiv_preserves_decision
RS2 DecisionQuotient.Information.rate_equals_srank
RS3 DecisionQuotient.Information.compression_below_srank_fails
RS4 DecisionQuotient.Information.srank_bits_sufficient
RS5 DecisionQuotient.Information.rate_distortion_bridge
S2P1 DecisionQuotient.Sigma2PHardness.exactlyIdentifiesRelevant_iff_sufficient_and_subset_relevantFinset
S2P2 DecisionQuotient.Sigma2PHardness.min_representationGap_zero_iff_relevant_card
S2P3 DecisionQuotient.Sigma2PHardness.min_sufficient_set_iff_relevant_card
S2P4 DecisionQuotient.Sigma2PHardness.representationGap
S2P5 DecisionQuotient.Sigma2PHardness.representationGap_eq_waste_plus_missing
S2P6 DecisionQuotient.Sigma2PHardness.representationGap_eq_zero_iff
S2P7 DecisionQuotient.Sigma2PHardness.representationGap_missing_eq_gapCard
S2P8 DecisionQuotient.Sigma2PHardness.representationGap_zero_iff_minimalSufficient
S2P9 DecisionQuotient.Sigma2PHardness.sufficient_iff_relevant_subset
SE1 DecisionQuotient.ClaimClosure.SE1
SE2 DecisionQuotient.ClaimClosure.SE2
SE3 DecisionQuotient.ClaimClosure.SE3
SE4 DecisionQuotient.ClaimClosure.SE4
SE5 DecisionQuotient.ClaimClosure.SE5
SE6 DecisionQuotient.ClaimClosure.SE6
SID1 ClaimClosure.dof1_zero_side_information
SID2 ClaimClosure.side_information_scales_with_redundancy
SOT1 ssot_iff
SR1 DecisionQuotient.ClaimClosure.SR1
SR2 DecisionQuotient.ClaimClosure.SR2
SR3 DecisionQuotient.ClaimClosure.SR3
SR4 DecisionQuotient.ClaimClosure.SR4
SR5 DecisionQuotient.ClaimClosure.SR5
TC1 DecisionQuotient.ToolCollapse.WorkProfile
TC2 DecisionQuotient.ToolCollapse.WorkModel
TC3 DecisionQuotient.ToolCollapse.ToolModel
TC4 DecisionQuotient.ToolCollapse.EventualStrictImprovement
TC5 DecisionQuotient.ToolCollapse.EffectiveCollapse
TC6 DecisionQuotient.ToolCollapse.tool_never_worse
TC7 DecisionQuotient.ToolCollapse.strict_improvement_has_witness
TC8 DecisionQuotient.ToolCollapse.effective_collapse_of_eventual_strict

Lean handle entry (continued)
TC9 DecisionQuotient.ToolCollapse.expBaseline
TC10 DecisionQuotient.ToolCollapse.linearTool
TC11 DecisionQuotient.ToolCollapse.linear_tool_eventual_strict
TC12 DecisionQuotient.ToolCollapse.linear_tool_effective_collapse
TL3 DecisionQuotient.ThermodynamicLift.joulesPerBit_pos_of_landauer_calibration
TL4 DecisionQuotient.ThermodynamicLift.energy_lower_mandatory_of_landauer_calibration
TUR1 DecisionQuotient.Physics.transitionProb_nonneg
TUR2 DecisionQuotient.Physics.transitionProb_sum_one
TUR5 DecisionQuotient.Physics.tur_bridge
TUR6 DecisionQuotient.Physics.multiple_futures_entropy_production
UO1 DecisionQuotient.UniverseDynamics
UO2 DecisionQuotient.feasibleActions
UO3 DecisionQuotient.lawDecisionProblem
UO4 DecisionQuotient.lawUtility
UO5 DecisionQuotient.logicallyDeterministic
UO6 DecisionQuotient.universe_sets_objective_schema
UO7 DecisionQuotient.lawUtility_eq_of_allowed_iff
UO8 DecisionQuotient.opt_eq_feasible_of_gap
UO9 DecisionQuotient.infeasible_not_optimal_of_gap
UO10 DecisionQuotient.opt_singleton_of_logicallyDeterministic
UO11 DecisionQuotient.opt_eq_of_allowed_iff
UQ1 DecisionQuotient.Physics.Uncertainty.binaryIdentityProblem
UQ2 DecisionQuotient.Physics.Uncertainty.binaryIdentityProblem_opt_true
UQ3 DecisionQuotient.Physics.Uncertainty.binaryIdentityProblem_opt_false
UQ4 DecisionQuotient.Physics.Uncertainty.exists_decision_problem_with_nontrivial_opt
UQ5 DecisionQuotient.Physics.Uncertainty.PhysicalNontrivialOptAssumption
UQ6 DecisionQuotient.Physics.Uncertainty.exists_decision_problem_with_nontrivial_opt_of_physical
W1 DecisionQuotient.Physics.single_future_zero_cost
W2 DecisionQuotient.Physics.transportCost_pos_of_offDiag
W3 DecisionQuotient.Physics.integrity_is_centroid
W4 DecisionQuotient.Physics.wasserstein_bridge
WD1 DecisionQuotient.checking_witnessing_duality_budget
WD2 DecisionQuotient.no_sound_checker_below_witness_budget
WD3 DecisionQuotient.checking_time_ge_witness_budget
WD4 DecisionQuotient.witnessBudgetEmpty
WD5 DecisionQuotient.checkingBudgetPairs
XC1 DecisionQuotient.Physics.srank_determines_states
XC2 DecisionQuotient.Physics.more_states_more_transport
XC3 DecisionQuotient.Physics.transport_lower_bound
XC4 DecisionQuotient.Physics.transport_independent_of_energy
XC5 DecisionQuotient.Physics.transport_independent_of_precision
XC6 DecisionQuotient.Physics.srank_unified_complexity
XC7 DecisionQuotient.Physics.complete_bridge_set

11.2 What Is Machine-Checked

The Lean formalization establishes:

1. **STOCHASTIC-SUFFICIENCY PP-completeness**: Reduction from MAJSAT showing stochastic sufficiency checking is PP-complete.
2. **SEQUENTIAL-SUFFICIENCY PSPACE-completeness**: Reduction from TQBF showing sequential sufficiency checking is PSPACE-complete.
3. **Substrate independence theorems**: Formal proofs that the matrix verdict is κ-independent while trajectories are κ-dependent.
4. **Bridge transfer conditions**: Formalized conditions under which sufficiency transfers between regimes.

11.3 Assumption Ledger (Auto)

Source files.

- `Paper4bStochasticSequential/ClaimClosure.lean`

Assumption bundles and fields.

- (No assumption bundles parsed.)

Conditional closure handles.

- (No conditional theorem handles parsed.)

First-principles Bayes derivation handles.

- (No Bayes-derivation theorem handles found.)

Compact Bayes handle IDs.

- (No compact Bayes alias IDs found.)

11.4 Complete Claim Coverage Matrix

Paper handle	Status	Lean support
`cor:exact-identifiability`	Derived	`DQ.Sigma2PHardness.exactlyIdentifiesRelevant_iff_sufficient_and_subset_relevantFinset`
`cor:exact-no-competence-zero-certified`	Full	`DQ.IntegrityCompetence.certificationOverheadBits_of_no_evidence`
`cor:gap-externalization`	Full	`DQ.HardnessDistribution.hardnessLowerBound`, `DQ.HardnessDistribution.right_dominates_wrong`
`cor:gap-minimization-hard`	Derived	`DQ.Sigma2PHardness.min_representationGap_zero_iff_relevant_card`
`cor:generalized-eventual-dominance`	Full	`DQ.HardnessDistribution.generalized_right_eventually_dominates_wrong`
`cor:information-barrier-query`	Full	`DQ.ClaimClosure.exact_admissible_iff_raw_lt_certified_total_core`, `DQ.ClaimClosure.exact_certainty_inflation_under_hardness_core`, `DQ.ClaimClosure.exact_raw_eq_certified_iff_certainty_inflation_core`
`cor:integrity-universal`	Full	`DQ.ClaimClosure.explicit_assumptions_required_of_not_excused_core`

cor:linear-positive-no-saturation	Full	DQ.HardnessDistribution.no_positive_slope_linear_represents_saturating
cor:no-auto-minimize	Full	DQ.ClaimClosure.explicit_state_upper_core
cor:no-uncertified-exact-claim	Full	DQ.ClaimClosure.information_barrier_opt_oracle_core
cor:outside-excuses-no-exact-report	Full	DQ.ClaimClosure.horizonTwoWitness_immediate_empty_sufficient
cor:overmodel-diagnostic-implication	Derived	DQ.Sigma2PHardness.representationGap_eq_zero_iff
cor:physics-no-universal-exact-claim	Full	DQ.ClaimClosure.hard_family_all_coords_core
cor:practice-bounded	Derived	DQ.ClaimClosure.tractable_bounded_core
cor:practice-diagnostic	Derived	DQ.Sigma2PHardness.min_representationGap_zero_iff_relevant_card
cor:practice-structured	Derived	DQ.ClaimClosure.tractable_separable_core
cor:practice-tree	Derived	DQ.ClaimClosure.tractable_tree_core
cor:practice-unstructured	Derived	DQ.ClaimClosure.hard_family_all_coords_core
cor:query-obstruction-bool	Full	DQ.ClaimClosure.minsuff_collapse_to_conp_conditional
cor:right-wrong-hardness	Full	DQ.HardnessDistribution.requiredWork
cor:type-system-threshold	Full	DQ.HardnessDistribution.native_dominates_manual
prop:abstention-frontier	Full	DQ.ClaimClosure.no_auto_minimize_of_p_neq_conp
prop:adq-ordering	Full	DQ.ClaimClosure.adq_ordering
prop:attempted-competence-matrix	Full	DQ.IntegrityCompetence.EvidenceForReport, DQ.IntegrityCompetence.ExactCertaintyInflation, DQ.IntegrityCompetence.certaintyInflation_iff_not_admissible
prop:bridge-failure-horizon	Full	DQ.ClaimClosure.declared_physics_no_universal_exact_certifier_core, DQ.ClaimClosure.dichotomy_conditional
prop:bridge-failure-stochastic	Full	DQ.ClaimClosure.posed_anchor_exact_claim_admissible_iff_competent
prop:bridge-failure-transition	Full	DQ.ClaimClosure.tractable_separable_core
prop:bridge-transfer-scope	Full	DQ.ClaimClosure.information_barrier_state_batch_core
prop:budgeted-crossover	Full	DQ.ClaimClosure.integrity_universal_applicability_core, DQ.PhysicalBudgetCrossover.SuccinctInfeasible
prop:certified-confidence-gate	Full	DQ.IntegrityCompetence.certificationOverheadBits, DQ.IntegrityCompetence.certificationOverheadBits_of_evidence
prop:crossover-above-cap	Full	DQ.ClaimClosure.integrity_resource_bound_for_sufficiency, DQ.PhysicalBudgetCrossover.SuccinctUnbounded
prop:crossover-not-certification	Full	DQ.ClaimClosure.meta_coordinate_irrelevant_of_invariance_on_declared_slice
prop:crossover-policy	Full	DQ.ClaimClosure.meta_coordinate_not_relevant_on_declared_slice
prop:declared-contract-selection-validity	Full	DQ.ClaimClosure.cost_asymmetry_eth_conditional, DQ.ClaimClosure.horizonTwoWitness_immediate_empty_sufficient, DQ.ClaimClosure.tractable_tree_core
prop:dominance-modes	Full	DQ.HardnessDistribution.centralized_higher_leverage
prop:empty-sufficient-constant	Full	DQ.DecisionProblem.relevantSet_is_minimal
prop:evidence-admissibility-equivalence	Full	DQ.IntegrityCompetence.ReportSignal
prop:exact-requires-evidence	Full	DQ.IntegrityCompetence.ReportBitModel, DQ.IntegrityCompetence.SignalConsistent
prop:generalized-assumption-boundary	Full	DQ.HardnessDistribution.generalized_dominance_can_fail_without_right_boundedness, DQ.HardnessDistribution.generalized_dominance_can_fail_without_wrong_growth
prop:hardness-conservation	Full	DQ.HardnessDistribution.simplicityTax_grows
prop:hardness-efficiency-interpretation	Full	DQ.HardnessDistribution.hardnessEfficiency_eq_central_share
prop:heuristic-reusability	Full	DQ.ClaimClosure.anchor_query_relation_false_iff, DQ.ClaimClosure.no_uncertified_exact_claim_core, DQ.ClaimClosure.oracle_lattice_transfer_as_regime_simulation, DQ.ClaimClosure.tractable_subcases_conditional
prop:identifiability-convergence	Full	DQ.ClaimClosure.epsilon_admissible_iff_raw_lt_certified_total_core
prop:insufficiency-counterexample	Full	DQ.ClaimClosure.DecisionProblem.sufficient_iff_zeroEpsilonSufficient, DQ.DecisionProblem.minimalSufficient_iff_relevant
prop:integrity-competence-separation	Full	DQ.IntegrityCompetence.Percent, DQ.IntegrityCompetence.admissible_matrix_counts

prop: integrity-resource-bound	Full	DQ.ClaimClosure.exact_raw_only_of_no_exact_admissible_core, DQ.IntegrityCompetence.abstain_signal_exists_with_guess_self, DQ.IntegrityCompetence.admissible_irrational_strictly_more_than_rational
prop:mdp-tractable	Full	DQ.ClaimClosure.process_bridge_failure_witness
prop: minimal-relevant-equiv	Full	DQ.ClaimClosure.DP6, DQ.ClaimClosure.DecisionProblem.epsOpt_zero_eq_opt
prop: no-evidence-zero-certified	Full	DQ.IntegrityCompetence.certifiedTotalBits
prop:one-step-bridge	Full	DQ.ClaimClosure.information_barrier_state_batch_core
prop:oracle-lattice-strict	Derived	DQ.ClaimClosure.horizonTwoWitness_immediate_empty_sufficient, DQ.ClaimClosure.information_barrier_opt_oracle_core
prop: oracle-lattice-transfer	Full	DQ.ClaimClosure.information_barrier_value_entry_core
prop:outside-excuses-explicit-assumptions	Full	DQ.ClaimClosure.cost_asymmetry_eth_conditional
prop: physics-no-universal-exact	Full	DQ.ClaimClosure.bridge_transfer_iff_one_step_class
prop:query-finite-state-generalization	Derived	DQ.ClaimClosure.process_bridge_failure_witness
prop:query-randomized-robustness	Derived	DQ.ClaimClosure.process_bridge_failure_witness
prop: query-randomized-weighted	Derived	DQ.ClaimClosure.process_bridge_failure_witness
prop: query-regime-obstruction	Full	DQ.ClaimClosure.minsuff_conp_complete_conditional
prop:query-state-batch-lb	Derived	DQ.ClaimClosure.horizon_gt_one_bridge_can_fail_on_sufficiency, DQ.ClaimClosure.process_bridge_failure_witness
prop: query-subproblem-transfer	Full	DQ.ClaimClosure.no_exact_identifier_implies_not_boundary_characterized, DQ.ClaimClosure.query_obstruction_finite_state_core, DQ.ClaimClosure.regime_core_claim_proved
prop: query-tightness-full-scan	Derived	DQ.ClaimClosure.process_bridge_failure_witness
prop:query-value-entry-lb	Derived	DQ.ClaimClosure.information_barrier_opt_oracle_core, DQ.ClaimClosure.process_bridge_failure_witness
prop: query-weighted-transfer	Derived	DQ.ClaimClosure.process_bridge_failure_witness
prop: refinement-strengthens	Full	DQ.ClaimClosure.stochastic_objective_bridge_can_fail_on_sufficiency
prop:retraction-evidence-integrity	Full	DQ.ClaimClosure.subproblem_hardness_lifts_to_full
prop:retraction-no-evidence-violates	Full	DQ.ClaimClosure.subproblem_transfer_as_regime_simulation
prop:selector-separation	Full	DQ.ClaimClosure.one_step_bridge
prop: sequential-bounded-horizon	Full	DQ.ClaimClosure.physical_crossover_policy_core
prop: sequential-static-relation	Full	DQ.ClaimClosure.physical_crossover_above_cap_core
prop:set-to-selector	Full	DQ.ClaimClosure.DP8
prop: snapshot-process-typing	Full	DQ.ClaimClosure.minsuff_collapse_core, DQ.ClaimClosure.poseAnchorQuery, DQ.ClaimClosure.system_transfer_licensed_iff_snapshot
prop: static-stochastic-strict	Derived	DQ.StochasticSequential.static_stochastic_strict_separation
prop: static-stochastic-transfer	Full	DQ.ClaimClosure.pose_returns_anchor_query_object
prop: stochastic-bounded-support	Full	DQ.ClaimClosure.posed_anchor_signal_positive_certified_implies_admissible
prop:stochastic-product-tractable	Full	DQ.ClaimClosure.query_obstruction_boolean_corollary
prop:stochastic-sequential-bridge-fail	Full	DQ.ClaimClosure.posed_anchor_no_competence_no_exact_claim
prop:stochastic-sequential-strict	Derived	DQ.StochasticSequential.stochastic_sequential_strict_separation
prop:sufficiency-char	Full	DQ.ClaimClosure.separable_detectable, DQ.ClaimClosure.snapshot_vs_process_typed_boundary
prop:thermo-conservation-additive	Full	DQ.ClaimClosure.sufficiency_conp_complete_conditional

prop: thermo-hardness-bundle	Full	DQ.ClaimClosure.sufficiency_iff_projectedOptCover_eq_opt
prop:thermo-lift	Full	DQ.ClaimClosure.sufficiency_conp_reduction_core, DQ.ClaimClosure.sufficiency_iff_dq_ratio
prop:thermo-mandatory-cost	Full	DQ.ClaimClosure.thermo_conservation_additive_core
prop: typed-claim-admissibility	Full	DQ.ClaimClosure.tractable_tree_core
prop: universal-solver-framing	Full	DQ.ClaimClosure.tree_structure_detectable
prop: zero-epsilon-competence	Full	DQ.IntegrityCompetence.RLFFWeights, DQ.IntegrityCompetence.certifiedTotalBits_ge_raw
prop: zero-epsilon-reduction	Full	DQ.ClaimClosure.DP7, DQ.DecisionProblem.sufficient_implies_selectorSufficient
thm:amortization	Full	DQ.HardnessDistribution.complete_model_dominates_after_threshold
thm:bridge-boundary-represented	Full	DQ.ClaimClosure.anchor_query_relation_true_iff, DQ.ClaimClosure.boundaryCharacterized_iff_exists_sufficient_subset, DQ.ClaimClosure.bounded_actions_detectable
thm: centralization-dominance	Full	DQ.HardnessDistribution.centralization_dominance_bundle, DQ.HardnessDistribution.centralization_step_saves_n_minus_one
thm:claim-integrity-meta	Full	DQ.ClaimClosure.cost_asymmetry_eth_conditional
thm:config-reduction	Full	DQ.ConfigReduction.config_sufficiency_iff_behavior_preserving
thm:cost-asymmetry-eth	Full	DQ.ClaimClosure.bridge_boundary_represented_family, DQ.HardnessDistribution.linear_lt_exponential_plus_constant_eventually
thm:dichotomy	Full	DQ.ClaimClosure.certified_total_bits_split_core, DQ.ClaimClosure.declaredBudgetSlice, DQ.ClaimClosure.declaredRegimeFamily_complete
thm:generalized-dominance	Full	DQ.HardnessDistribution.generalized_right_dominates_wrong_of_bounded_vs_identity_lower
thm:generalized-saturation-possible	Full	DQ.HardnessDistribution.generalizedTotal_with_saturation_eventually_constant, DQ.HardnessDistribution.requiredWork_eq_affine_in_sites
thm: linear-saturation-iff-zero	Full	DQ.HardnessDistribution.saturatingSiteCost_eventually_constant
thm:overmodel-diagnostic	Full	DQ.ClaimClosure.anchor_sigma2p_reduction_core, DQ.ClaimClosure.horizon_gt_one_bridge_can_fail_on_sufficiency
thm:regime-coverage	Full	DQ.ClaimClosure.bridge_failure_witness_non_one_step, DQ.ClaimClosure.no_exact_claim_admissible_under_hardness_core
thm:tax-conservation	Full	DQ.HardnessDistribution.gap_conservation_card
thm:tax-grows	Full	DQ.HardnessDistribution.hardnessLowerBound
thm:tractable	Full	DQ.ClaimClosure.thermo_eventual_lift_core, DQ.ClaimClosure.thermo_hardness_bundle_core, DQ.ClaimClosure.thermo_mandatory_cost_core, DQ.ClaimClosure.tractable_bounded_core
thm: typed-completeness-static	Full	DQ.ClaimClosure.transition_coupled_bridge_can_fail_on_sufficiency

Notes: (1) Full rows come from theorem-local inline anchors in this paper. (2) Derived rows are filled by dependency/scaffold claim-handle derivation (same paper-handle label across proof dependencies). (3) Unmapped means no local anchor and no derivable dependency support were found.
Auto summary: mapped 102/102 (full=84, derived=18, unmapped=0).

11.5 Claims Not Fully Mechanized

Entries marked **Full (conditional)** are explicitly mechanized transfer theorems that depend on standard external complexity facts (e.g., MAJSAT PP-completeness, TQBF PSPACE-completeness), with those dependencies represented as hypotheses in Lean.

11.6 Verification

The proofs compile with Lean 4. Run `lake build` in the proof directory to verify.

References

[1] Edoardo Amaldi and Viggo Kann. On the approximability of some maximum spanning tree problems. *Theoretical Computer Science*, 196(1-2):3–19, 1998.

[2] Avrim L. Blum and Pat Langley. Selection of relevant features and examples in machine learning. *Artificial Intelligence*, 97(1-2):245–271, 1997.

[3] Ronald A. Fisher. On the mathematical foundations of theoretical statistics. *Philosophical Transactions of the Royal Society of London. Series A*, 222:309–368, 1922.

[4] Isabelle Guyon and André Elisseeff. An introduction to variable and feature selection. *Journal of Machine Learning Research*, 3:1157–1182, 2003.

[5] Ronald A. Howard. Information value theory. *IEEE Transactions on Systems Science and Cybernetics*, 2(1):22–26, 1966.

[6] Michael L. Littman, Judy Goldsmith, and Martin Mundhenk. The computational complexity of probabilistic planning. *Journal of Artificial Intelligence Research*, 9:1–36, 1998.

[7] Paul Milgrom and John Roberts. Price and advertising signals of product quality. *Journal of Political Economy*, 94(4):796–821, 1986.

[8] Martin Mundhenk, Judy Goldsmith, Christopher Lusena, and Eric Allender. The complexity of finite-horizon markov decision process problems. *Journal of the ACM*, 47(4):681–720, 2000.

[9] Roger B. Myerson. Incentive compatibility and the bargaining problem. *Econometrica*, 47(1):61–73, 1979.

[10] Christos H. Papadimitriou and John N. Tsitsiklis. The complexity of markov decision processes. *Mathematics of Operations Research*, 12(3):441–450, 1987.

[11] Howard Raiffa and Robert Schlaifer. *Applied Statistical Decision Theory.* Division of Research, Graduate School of Business Administration, Harvard University, Boston, MA, 1961. Library records list this imprint; avoid using JSTOR DOI 10.2307/2282452 which refers to a journal item, not the book.

[12] Michael Spence. Job market signaling. *The Quarterly Journal of Economics*, 87(3):355–374, 1973.